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Automated Solar Panel Fault Detection Using Hybrid Wavelet Transformation and Deep Learning Networks

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Declaration

I hereby declare that this thesis, titled "Automated Solar Panel Fault Detection Using Hybrid Wavelet Transformation and Deep Learning Networks", is my own work carried out under the supervision of Dr. Md. Salah Uddin Yusuf, Professor, Department of Electrical and Electronic Engineering, Khulna University of Engineering & Technology. The work presented here has not been submitted elsewhere for the award of any other degree, and all sources of information have been duly acknowledged.

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Abstract

Automated inspection of photovoltaic modules from infrared thermal images is now essential for maintaining large solar installations, where hundreds of thousands of modules must be screened for faults. This thesis studies whether a band selective hybrid wavelet enhancement improves the automated classification of module faults on the InfraredSolarModules dataset, which contains 20,000 low resolution thermal images across twelve classes with a severe class imbalance. The question is examined under a statistically controlled protocol that is rarely applied in this field. A corrected band selective enhancement is developed, in which contrast operations act only on the wavelet approximation band and sharpening acts only on the detail bands. Using EfficientNetV2 and a multi seed paired test, the enhancement is shown not to reliably improve classification when used as a direct replacement for the raw image, and to be significantly gives worse output on macro averaged F1. Taking this negative result seriously, the enhancement is instead used as a complementary feature. The features of the raw and enhanced views are fused, the class imbalance is corrected with SMOTE, the backbone is fine tuned on the thermal images, and a second backbone, ConvNeXt, is added in a weighted ensemble alongside EfficientNetV2. On the binary task of separating healthy from faulty modules, the system reaches 0.930 accuracy and an area under the curve of 0.977. On the harder twelve class task, it reaches 0.811 accuracy and 0.665 macro F1, with clear gains on the rare fault classes. The contribution of this thesis is both the method, the fusion based two backbone ensemble, and the lesson that a preprocessing method should be evaluated in a controlled setting and considered as a complement rather than a substitute.

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Nomenclature

AUC	Area Under the Receiver Operating Characteristic Curve
CLAHE	Contrast Limited Adaptive Histogram Equalisation
CNN	Convolutional Neural Network
CNR	Contrast to Noise Ratio
DWT	Discrete Wavelet Transform
F1	Harmonic Mean of Precision and Recall
GELU	Gaussian Error Linear Unit
Grad-CAM	Gradient Weighted Class Activation Mapping
HF	High Frequency
IDWT	Inverse Discrete Wavelet Transform
LL, LH, HL, HH	Wavelet Approximation and Detail Sub Bands
MBConv	Mobile Inverted Bottleneck Convolution
MLP	Multilayer Perceptron
PSNR	Peak Signal to Noise Ratio
PV	Photovoltaic
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
SMOTE	Synthetic Minority Over Sampling Technique
SSIM	Structural Similarity Index
β	CLAHE Clip Limit
γ	Gamma Correction Exponent
g_{hf}	High Frequency Gain

Chapter 1

Introduction

1.1 Background

Solar photovoltaic (PV) generation has become one of the fastest-growing sources of electricity worldwide, and its continued expansion is central to most national decarbonization strategies. Globally installed PV capacity had already reached several hundred gigawatts by the end of the last decade [2] and has continued to grow steeply since. This scale has an operational consequence that is easy to overlook: a modern utility-scale plant is not a single large machine but a field of tens or hundreds of thousands of nominally identical modules, each of which can fail independently. A plant in the tens of megawatts size can have over a hundred thousand modules, each of which can be damaged during transport, have a bypass diode fail after several years, accumulate soiling during a dry season, or otherwise lose power.

The faults that occur in fielded PV modules are diverse in both physical origin and electrical signature. Comprehensive reviews of PV module failure modes [3] distinguish manufacturing and handling defects such as cell cracking; electrical faults such as an activated bypass diode (which typically disables roughly a third of a module); localized thermal defects such as hot spots caused by cell mismatch or resistive faults; and environmental effects such as soiling, vegetation encroachment, and partial shading. The consequences span a wide range of severity, from a modest reduction in energy yield through accelerated degradation to, in the case of persistent hot spots, a genuine fire hazard. Because undetected faults directly erode both the energy yield and the safety margin of an installation, regular inspection is not optional but a routine maintenance requirement for any serious PV asset.

Since the large majority of these faults perturb the surface temperature Infrared (IR) thermography has become the preferred method of distribution of the impacted module. A faulty cell releases as heat the power that would otherwise be used to light the bulb, which is dissipated. A healthy cell would change to electricity and that distinction would be recorded as a distinct thermal signature. The main drawback of thermography is that it is difficult to understand from a distance and quick view. The early inspection was made using handheld devices. Imagery has advanced ground up, and the field has firmly shifted to aerial, Infrared thermography, or an unmanned aerial vehicle with a thermal camera. Sweeps whole rows of plants. Tools of this kind have been shown to enable the inspection of several megawatts of installed capacity per day, with the potential to scale to far larger plants [2]. The bottleneck has therefore shifted away from image acquisition and toward interpretation. Meanwhile, somebody, or something, must examine every module crop and decide whether it is healthy and, if not, which kind of fault types it exhibits.

Manual interpretation does not scale to this volume of imagery. It is slow, it requires trained inspectors, and for the rarer fault categories, it is also inconsistent, because a human reviewer may encounter a given rare class only a handful of times in a year. These limitations have motivated a large and active body of research into automating PV fault recognition with machine learning and, in particular, with deep convolutional neural networks (CNNs), which now dominate the field. Recent reviews report that deep learning methods consistently outperform classical machine-learning pipelines for thermographic PV inspection [4, 5], and individual studies using CNN architectures on infrared data have reported fault-classification accuracies well above 90% under favourable conditions [6]. Grad-CAM and related explainability techniques are increasingly used alongside these models to confirm that predictions are genuine defect regions rather than spurious image cues.

It is important to read such headline accuracies in context, however, because they depend strongly on the dataset and the evaluation protocol. A recurring and well-documented characteristic of real PV anomaly data is severe class imbalance. The overwhelming majority of modules in any working plant are healthy, so healthy examples vastly outnumber any individual fault type. Work that explicitly accounts for realistic conditions, for example, by testing on a different plant from the one used for training or by reporting metrics that are not dominated by the healthy majority class, typically reports more modest figures. Bommes et al., training on millions of aerial IR images under a cross-plant protocol, reported area-under-curve scores ranging from 73.3% to 96.6% depending on the source and target plants [7], illustrating how much performance depends on the difficulty of the evaluation setting rather than on the model alone.

The dataset used throughout this thesis is InfraredSolarModules, released by Raptor Maps and now widely adopted as a benchmark for this problem [8]. It contains 20,000 thermal crops of individual PV modules, each labelled into one of 12 classes: eleven anomaly types together with a No-Anomaly (healthy) class. The authors of the dataset deliberately preserved its natural class imbalance precisely because that imbalance is a defining feature of the real problem. Two properties of the dataset shape every experiment reported in this thesis. The first is resolution: each image is natively only 24×40 pixels, a severe constraint on how much fine detail any algorithm can recover. The second is the imbalance itself: the No-Anomaly class accounts for roughly half of all images, while the rarest fault class contains fewer than 180 examples, a ratio in excess of fifty to one. Table 1.1 lists the full distribution.

Table 1.1: Class distribution of the InfraredSolarModules dataset.

Class	**Images**	**Class**	**Images**
No-Anomaly	10,000	Cracking	941
Cell	1,877	Offline-Module	826
Vegetation	1,639	Hot-Spot	251
Diode	1,499	Hot-Spot-Multi	247
Cell-Multi	1,288	Soiling	205
Shadowing	1,056	Diode-Multi	175

The combination of low resolution and low contrast is what makes image enhancement an attractive preprocessing step for this particular problem. If a defect mark occupies only a handful of pixels and differs from its surroundings by a small number of intensity levels, then improving local contrast before classification is a reasonable strategy. Wavelet-domain enhancement offers a principled framework for doing so, because the discrete wavelet transform (DWT) decomposes an image into frequency sub-bands that can be processed independently: coarse contrast and brightness information concentrate in the low-frequency approximation band, while edges and fine texture occupy the high-frequency detail bands. Restricting contrast operations to the approximation band, rather than applying them indiscriminately across the whole image, is a well-established way to avoid the noise amplification that bothers naive contrast-limited adaptive histogram equalisation (CLAHE) [9]. This thesis investigates whether such an enhancement step, carefully designed, actually improves automated fault classification on low-resolution thermal PV imagery, and does so under an evaluation protocol strict enough to trust the answer.

1.2 Motivation

The motivation for this thesis has two components, one technical and one methodological. The technical component arose directly from a failure encountered in an earlier version of this work.

The first enhancement pipeline implemented in this project applied CLAHE, gamma correction, and wavelet high-frequency sharpening in sequence, each operating over the entire reconstructed image. This seemed a reasonable design: every operation is individually well established in the image-enhancement literature, so composing them ought to compound their benefits. An ablation study comparing the full cascade against its individual components demonstrated the opposite. The complete pipeline scored worse on both contrast-to-noise ratio (CNR) and structural similarity (SSIM) than any single component applied in isolation. The operations were interfering with one another, the local contrast stretching introduced by CLAHE was being amplified a second time by the subsequent global sharpening pass over the same pixels, degrading the image rather than clarifying it.

That failure motivated the corrected design that this thesis investigates: a band-selective hybrid wavelet enhancement, in which the contrast operations (CLAHE and gamma correction) are confined to the low-frequency approximation band while the sharpening operation is confined to the high-frequency detail bands. Under this design the two mechanisms act on disjoint frequency content and therefore cannot compound each other's side effects, which is precisely the failure mode that undermined the original cascade.

The methodological component of the motivation is, in retrospect, the more important between the two. A recurring pattern in the applied literature on this problem is that a preprocessing or architectural modification is proposed, a model is trained once with it and once without it, and the difference between the two runs is presented as evidence that the modification is beneficial. The difficulty with this protocol is that fine-tuning a large convolutional network on a small, heavily imbalanced dataset is a noisy process. Weight initialization, batch ordering, and hardware non-determinism all contribute to run-to-run variance, and if that variance is large enough, then a single-run comparison cannot distinguish a genuine improvement from chance.

This concern was not hypothetical in the present work. Two independent training runs of the same enhanced-input configuration produced test accuracies of 53.6% and 42.9%, a spread substantially larger than the gap between the enhanced and baseline models within either individual run. Had only the first run been performed, the enhancement would have appeared to deliver a clear improvement of several percentage points; had only the second been performed, it would have appeared to degrade. Neither conclusion would have been trustworthy. This observation shaped the entire evaluation strategy of the thesis, which relies on repeated runs and a paired statistical test rather than on any single training result.

The main question we want to answer is this: Can a special step that improves the image called wavelet enhancement really help us better identify problems, in pictures of PV imagery? We want to make sure this step is done correctly and that we are looking at the things. If this step does not help that means it is not useful or maybe we are just not using it the way. We need to figure out if the wavelet enhancement step's helpful or not. Does it actually make a difference when we look at PV imagery?

1.3 Objectives

Building on this motivation, the specific objectives of this thesis are as follows.

1. To design and implement a band-selective hybrid wavelet enhancement pipeline for low-resolution thermal PV imagery, combining DWT decomposition, CLAHE and gamma correction restricted to the approximation sub-band, and high-frequency gain restricted to the detail sub-bands, followed by inverse DWT reconstruction.

2. To quantify the effect of this pipeline on image quality at the pixel level, using CNR, SSIM, and peak signal-to-noise ratio (PSNR) computed on a per-class basis, rather than assuming that a single globally tuned parameter set benefits all twelve fault categories equally.
3. To train an EfficientNetV2-S classifier [1] on enhanced images and, as a controlled ablation, on raw resized images, holding the architecture, data split, training schedule, and class-imbalance handling identical between the two configurations so that the enhancement step is the only variable under study.
4. To benchmark the proposed wavelet enhancement against an alternative enhancement strategy, so that the comparison is not merely enhancement versus no enhancement but enhancement against a competing method.
5. To verify, using Grad-CAM [10], that the trained classifier bases its predictions on physically meaningful defect regions rather than on background or framing artefacts.
6. To investigate whether the raw and enhanced representations can be combined more productively than by substitution, specifically, by fusing pooled convolutional features extracted from both representations and correcting the residual class imbalance with the synthetic minority over-sampling technique (SMOTE) [11] ensembling.

It is worth stating plainly at the outset that the central experimental finding of this thesis is a negative one with respect to the original hypothesis: used as a direct substitute for the raw image, the enhancement does not reliably improve classification performance, and by the most imbalance-sensitive metric it mildly harms it. The value of the work lies in having established that result under conditions strict enough to trust, and in the observation, which follows from taking the negative result seriously rather than working around it, that the same enhancement does contribute usefully when it is treated as a complementary feature source rather than a replacement.

1.4 Organization of the Thesis

The remainder of this thesis is organised as follows. **Chapter 2** reviews the relevant literature: deep-learning approaches to PV fault classification from thermal imagery and the performance levels they report on this benchmark, wavelet- and CLAHE-based image enhancement, the EfficientNetV2 architecture family, techniques for handling class imbalance, and gradient-based explainability methods, closing by identifying the specific gap that this thesis addresses. **Chapter 3** sets out the methodology in full: the dataset and splitting protocol, the four-stage band-selective enhancement algorithm together with its governing equations, the classifier architecture and two-phase transfer learning schedule, the class-imbalance handling strategy, the evaluation metrics, the multi-seed statistical validation procedure, the Grad-CAM implementation, and the feature-fusion and SMOTE pipeline. **Chapter 4** presents and discusses the experimental results: per-class enhancement quality, the enhanced-versus-raw classification comparison across independent runs, the multi-seed significance test that resolves the disagreement between those runs, the tuned oversampling run, the comparison against homomorphic filtering, the Grad-CAM visualizations, and the feature-fusion results that constitute the best-performing configuration. **Chapter 5** relates the project to the real-world engineering problem that motivates it, the engineering activities undertaken, its socio-economic and environmental impacts, and its mapping to the relevant United Nations Sustainable Development Goals. **Chapter 6** summarizes the findings, states the limitations of the study candidly, and sets out directions for future work.

Chapter 2

Literature Review

This chapter surveys the work on which this thesis builds. Section 2.1 examines automated photovoltaic fault detection from thermal imagery and, in particular, the performance levels reported on the benchmark dataset used here. Section 2.2 reviews image enhancement in the wavelet domain, which provides the theoretical foundation for the proposed method. Section 2.3 discusses convolutional networks and the classifier backbone, Section 2.4 the treatment of class imbalance, and Section 2.5 feature level fusion. Section 2.6 reviews the classical classifiers and ensemble methods applied over deep features. Section 2.7 then addresses a concern that is central to this thesis but comparatively neglected in the applied literature: the statistical reliability of single run experimental comparisons. Section 2.8 covers explainability, and Section 2.9 synthesises the review into a statement of the research gap.

2.1 Automated PV Fault Detection from Thermal Imagery

2.1.1 Failure Modes and the Case for Thermography

Photovoltaic modules fail in a variety of ways, and the classification taxonomy used by any automated system ultimately derives from the underlying physics of those failures. The comprehensive review by Akram et al. [3] organises PV module failures by origin, namely manufacturing defects, installation damage, and degradation accumulated during operation, and observes that a large proportion of them alter the thermal behaviour of the affected cell or module. This is the physical basis for infrared inspection: a defective cell converts less of the incident irradiance to electrical power and dissipates the remainder as heat, producing a localised temperature elevation that a thermal camera can resolve.

The practical value of this observation grew substantially with the adoption of unmanned aerial vehicles for inspection. Bommes et al. [2] developed a computer-vision pipeline for semi-automatic extraction of individual PV modules from thermographic UAV video, using it to assemble a dataset of 4.3 million infrared images covering 107,842 modules across seven plants. They report that their tool enables inspection of between 3.5 and 9 MW_p of installed capacity per day, and that a ResNet-50 trained on the resulting data classified ten anomaly types with more than 90% test accuracy. Work of this scale demonstrates both the feasibility of automated aerial inspection and the volume of imagery it generates, a volume that makes manual interpretation impractical and automation necessary.

2.1.2 Deep Learning Approaches and Reported Performance

Recent reviews agree that deep learning has displaced classical machine learning as the dominant approach for thermographic PV diagnosis. Masita et al. [4] survey deep learning defect detection for PV modules across a broad range of architectures and datasets, and Boubaker et al. [5] report, from a direct comparative assessment, that deep models consistently outperform conventional machine learning pipelines on infrared PV data. Individual studies using convolutional architectures have reported high accuracies: for example, Bu et al. [6] developed a CNN-based classification method for thermographic PV cell images achieving above 97% accuracy, and earlier work by Dunderdale et al. [12] established the viability of machine learning classification of PV defects from thermal infrared imaging.

Two studies are particularly relevant to the present thesis because they operate on the same public dataset. Millendorf et al. [8] introduced the InfraredSolarModules dataset itself, comprising 20,000 images of 24×40 pixels across 12 classes, and were explicit that the dataset was constructed to highlight the class imbalance characteristic of real PV anomaly data. Duranay [13] subsequently reported strong results on this dataset using a hybrid strategy: deep features were extracted from a pretrained EfficientNet-B0, reduced by neighbourhood component analysis (NCA) feature selection, and classified by a support vector machine under ten-fold cross-validation, yielding 93.9% accuracy and an 89.8% macro F1-score. This result is important for two reasons. First, it establishes a strong reference point on the identical benchmark. Second, and more significantly for the design of this thesis, it demonstrates that a feature extraction plus classical classifier pipeline can outperform naive end to end fine tuning on this dataset, a pattern that recurs in the fusion experiments reported in Chapter 4.

2.1.3 Interpreting Reported Accuracies

Headline accuracies in this literature must be read with attention to the evaluation protocol, because the difficulty of the evaluation setting affects results at least as much as the model architecture does. Bommes et al. [7] make this point directly. Observing that most published work samples training and test data from the same distribution, they reframe PV fault detection as an unsupervised domain adaptation problem in which a model trained on one plant must generalise to a different plant. Under this more realistic protocol, and using a ResNet-34 trained with a supervised contrastive loss followed by a k-nearest-neighbour classifier, they report areas under the receiver operating characteristic curve ranging from 73.3% to 96.6% across nine source–target plant combinations. The width of that range, obtained by a single well designed method, is a caution against comparing accuracies across studies without also comparing their evaluation conditions.

The metric itself matters equally. On a dataset where a single healthy class constitutes half of all samples, overall accuracy can be high while performance on rare fault classes remains poor. Macro-averaged F1, which computes the metric independently per class and averages without weighting by class frequency, is substantially more demanding, and it is the primary metric adopted throughout this thesis for that reason.

2.1.4 Binary Detection versus Multiclass Diagnosis

A further distinction runs through this literature, and it matters for how the present work is framed. Fault recognition can be posed at two levels of difficulty. The first is binary detection, where the only question is whether a module is healthy or faulty. The second is multiclass diagnosis, where the specific fault type must also be named. These are genuinely different tasks, and the reported results reflect that: the binary problem is consistently easier, because collapsing every fault into a single faulty class removes the hard confusions between fault types that look alike in a thermal image.

Several studies report both levels on the same data and make the gap plain. Working under Algerian field conditions, Belhaouas et al. [14] reached 92.5% accuracy for binary defect detection but only 78.85% when the same images had to be sorted into eight fault types. A very recent hierarchical framework by Author and Author [15] follows the same logic deliberately, running a fast binary healthy versus defective stage first and only then separating the fault types, and it too reports markedly higher scores on the binary stage than on the eight class stage. On the specific dataset used here, binary healthy versus faulty variants have likewise been derived and shown to be easier than the full twelve class problem. Boubaker et al. [5] go as far as building their pipeline around a binary detector, on the reasoning that in practice the first thing an operator needs to know is simply which modules require attention.

This two-level view shapes the present thesis. Binary detection answers the safety critical question of whether a module is faulty at all, and it is the natural first result to establish; multiclass diagnosis answers the finer question of which fault is present, and it is correspondingly harder on low resolution, imbalanced data. Both are reported in Chapter 4, and the difference in difficulty between them is exactly what the literature above would predict.

2.2 Image Enhancement in the Wavelet Domain

2.2.1 Contrast Limited Adaptive Histogram Equalization

Histogram equalisation redistributes image intensities to occupy the available dynamic range more uniformly. Applied globally it is often too aggressive, and adaptive histogram equalisation, which operates on local tiles, tends to amplify noise in near-uniform regions. Zuiderveld [16] addressed this with contrast limited adaptive histogram equalization (CLAHE), which caps the local histogram at a clip limit β and redistributes the excess before computing the transformation. For a tile with histogram $h(i)$ over L intensity levels, the clipped histogram is

$$h_{\text{clip}}(i) = \min\big(h(i),\, \beta\big) + \frac{1}{L}\sum_{j=0}^{L-1} \max\big(h(j) - \beta,\, 0\big), \tag{2.1}$$

where the summed term redistributes clipped mass uniformly across all levels. The cumulative distribution of h_{clip} then defines the intensity mapping for that tile, with bilinear interpolation applied between tile centres to avoid visible block boundaries. The clip limit β directly governs the trade-off between contrast gain and noise amplification, and its selection is therefore consequential for low signal-to-noise imagery such as thermal PV data.

A complementary point operation frequently paired with CLAHE is gamma correction, which adjusts overall luminance through a non-linear power law. For a normalised intensity $I \in [0, 1]$,

$$I_\gamma = I^\gamma, \tag{2.2}$$

where $\gamma < 1$ brightens the image by expanding the contrast of darker regions and $\gamma > 1$ darkens it. Applied to the low-frequency band of a decomposition, gamma correction and CLAHE together provide control over both global luminance and local contrast while leaving high-frequency detail untouched.

2.2.2 Multiresolution Decomposition and the Discrete Wavelet Transform

The theoretical basis for wavelet-domain image processing is the multiresolution framework of Mallat [17], which shows that an image can be decomposed into a coarse approximation together with detail components at successive scales. For a two-dimensional image, one level of the discrete wavelet transform applies a low-pass filter h and a high-pass filter g separably along rows and columns, followed by dyadic downsampling, yielding four sub-bands:

$$\text{DWT2}(I) \;\longrightarrow\; \{\,\text{LL},\ \text{LH},\ \text{HL},\ \text{HH}\,\}, \tag{2.3}$$

where LL is the low-pass approximation, and LH, HL, and HH capture horizontal, vertical, and diagonal detail respectively. Explicitly, the sub-band coefficients at scale $j + 1$ are obtained from the approximation at scale j by

$$c_{j+1}^{\text{LL}}[m, n] = \sum_k \sum_l h[k - 2m]\, h[l - 2n]\, c_j[k, l], \tag{2.4}$$

with the three detail sub-bands following the same form under the substitutions (h, g), (g, h), and (g, g) for the row and column filters. The transform is invertible: applying the corresponding synthesis filters and upsampling reconstructs the original image exactly, so any modification made to the sub-band coefficients propagates to the reconstruction in a controlled way. Figure 2.1 illustrates the decomposition and the frequency content of each sub-band.

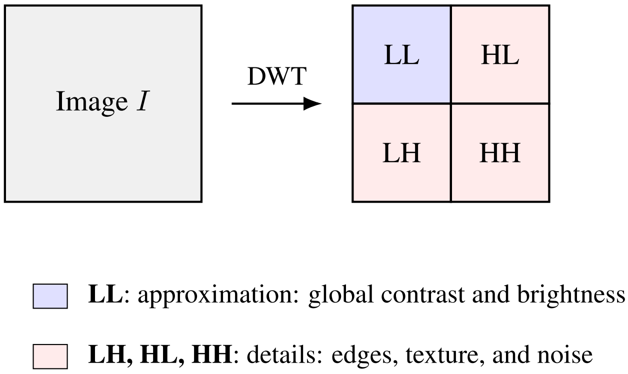

Figure 2.1: One-level two-dimensional discrete wavelet decomposition into the LL, LH, HL, and HH sub-bands.

2.2.3 Band-Selective Enhancement

The separation illustrated in Figure 2.1 has a direct practical consequence, and it is one that the enhancement literature has exploited consistently. If CLAHE is applied to the whole image, it operates on the noise present in the detail bands as well as the contrast information in the approximation band, and the clip limit must then be chosen as a compromise between the two. Applying it only to the LL sub-band removes that compromise.

Lidong et al. [9] formalised this combination, applying CLAHE to the low-frequency sub-band of a wavelet decomposition before reconstruction, and reported improved contrast with reduced noise amplification relative to direct CLAHE. Qi et al. [18] applied the same principle specifically to infrared imagery, enhancing the largest-scale low-frequency sub-band with CLAHE while filtering the high-frequency coefficients to suppress noise, and reported improvements in contrast, detail visibility, and overall visual quality on a large set of infrared images. Sahu et al. [19] likewise combined wavelet transforms with CLAHE applied to the low-frequency component only, and Yang et al. [20] developed a related scheme in which the low-frequency coefficients are processed by a homomorphic filter and the high-frequency coefficients by threshold shrinkage before reconstruction.

Several observations follow from this body of work and directly shaped the design evaluated in this thesis. The first is that band-selective application of CLAHE is not a novel contrivance but standard practice, motivated by the noise-amplification problem. The second is that the detail sub-bands are typically treated separately, either suppressed to reduce noise or amplified to sharpen edges, rather than left untouched. The third, and least frequently stated, is that these methods work because the operations are confined to disjoint sub-bands; composing multiple enhancement operations over the same full-resolution pixels forfeits that guarantee. The failure of the whole-image cascade described in Chapter 3 is a direct empirical demonstration of this last point.

2.2.4 Quantifying Enhancement Quality

Enhancement methods are conventionally assessed with complementary metrics that measure different properties. Contrast-to-noise ratio quantifies the separation between a region of interest and its background relative to the background noise:

$$\mathrm{CNR} = \frac{|\mu_f - \mu_b|}{\sigma_b}, \tag{2.5}$$

where μ_f and μ_b are the mean intensities of the foreground and background regions and σ_b is the background standard deviation. The structural similarity index of Wang et al. [21] instead measures perceptual similarity between the enhanced image and its original,

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \tag{2.6}$$

and peak signal-to-noise ratio (PSNR) measures fidelity to the original in a mean-squared-error sense,

$$\mathrm{PSNR} = 10 \cdot \log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}}\right), \qquad \mathrm{MSE} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}\big[I(i,j) - K(i,j)\big]^2, \tag{2.7}$$

where I and K are the original and processed images of size $M \times N$, MSE is the mean squared error between them, and MAX_I is the maximum possible pixel value (255 for 8-bit images). A higher PSNR indicates that the processed image departs less from the original.

These metrics do not measure the same thing, and this distinction matters for interpreting the enhancement comparison presented in Chapter 4. CNR rewards a method for changing the image in a way that increases defect contrast, whereas SSIM and PSNR reward a method for not changing it. A composite score that weights SSIM and PSNR heavily therefore implicitly favours conservative methods, and a method that aggressively improves defect contrast may score poorly on such a composite despite doing exactly what enhancement is intended to do. Interpreting these metrics as though they were interchangeable measures of enhancement quality is a straightforward way to reach a misleading conclusion.

2.3 Convolutional Networks, Backbones, and Transfer Learning

2.3.1 Convolutional Neural Networks for Image Classification

Convolutional neural networks (CNNs) are the standard tool for image classification and underpin essentially all of the PV fault-detection work discussed in Section 2.1. A CNN learns a hierarchy of features directly from pixels: early convolutional layers respond to low-level structure such as edges and corners, and deeper layers compose these into progressively more abstract and task-specific patterns, with pooling layers providing a degree of translation invariance and a final fully connected stage mapping the learned representation to class scores [22]. A recurring practical obstacle, particularly relevant to very deep networks, is that gradients can vanish or explode during training; residual networks addressed this with skip connections that allow gradients to bypass layers, enabling the reliable training of much deeper models [23]. The backbones used in modern PV inspection, namely ResNet in Bommes et al. [2] and Bommes et al. [7], EfficientNet in Duranay [13], and EfficientNetV2 in this thesis, all inherit from this line of development.

2.3.2 EfficientNet and Compound Scaling

Tan and Le [24] introduced the EfficientNet family, whose central contribution was compound scaling: rather than increasing network depth, width, or input resolution in isolation, all three are scaled together according to a fixed ratio determined by a small grid search. This yielded models that matched or exceeded the accuracy of much larger architectures at a fraction of the parameter count. Tan and Le [1] subsequently developed EfficientNetV2 using training-aware neural architecture search combined with progressive learning, in which image resolution and regularisation strength are increased jointly during training. The resulting models train several times faster than the original EfficientNet family and than comparably accurate vision transformers, while using fewer parameters.

Training speed was the decisive consideration in selecting EfficientNetV2-S as the backbone for this thesis. The experimental design requires the same architecture to be trained many times over, covering enhanced and raw configurations, three random seeds each, a tuned oversampling run, and separate feature extraction passes, and a backbone that trains slowly would have made the multi seed statistical validation described in Section 2.7 infeasible within the available compute budget.

2.3.3 Transfer Learning and Fine-Tuning

Training a deep CNN from random initialisation requires substantially more labelled data than the 20,000 low resolution images available here. The standard remedy is transfer learning:

initialise from weights pretrained on a large corpus such as ImageNet [25] and adapt them to the target task. Yosinski et al. [26] showed empirically that features learned in early convolutional layers are general and transfer well across tasks, whereas later layers are progressively more specific to the original task; this is the basis for the common practice of freezing early layers and adapting only the later ones.

Two regimes of transfer learning are relevant to this thesis. In feature extraction, the pretrained backbone is frozen entirely and used only to produce a fixed embedding, on top of which a separate classifier is trained; this is fast and data-efficient, and it is the regime in which the feature-fusion experiments of this thesis operate. In fine tuning, some or all of the backbone's weights are unfrozen and updated on the target data, typically at a low learning rate to avoid destroying the pretrained representation; this can adapt the features more closely to the target domain, here thermal rather than natural images, at the cost of greater compute and a higher risk of overfitting on a small dataset. The trade-off between these two regimes, and the fact that the feature extraction regime is used for the fusion experiments, is revisited in Chapter 3.

2.4 Class Imbalance

2.4.1 The Nature of the Problem

When one class dominates a training set, a classifier minimising overall error is rewarded for predicting that class, and rare classes are systematically under-predicted. Buda et al. [27] conducted a systematic study of this effect in convolutional networks, finding that imbalance consistently degrades performance and that its impact grows with the severity of the skew. Johnson and Khoshgoftaar [28] survey the deep learning-specific literature and organise the available remedies into data level methods that alter the training distribution, algorithm-level methods such as cost-sensitive loss weighting, and hybrid approaches.

For the InfraredSolarModules dataset the problem is acute: the healthy class outnumbers the rarest fault class by more than fifty to one. Because the practical purpose of the system is precisely to identify faults, and because some rare classes carry the greatest safety consequence, a model that achieves high accuracy by defaulting to the healthy prediction is of little operational value. This is the central argument for reporting macro-averaged F1 alongside accuracy. Figure 2.2 shows the class distribution on a logarithmic scale, which makes the severity of the imbalance immediately visible.

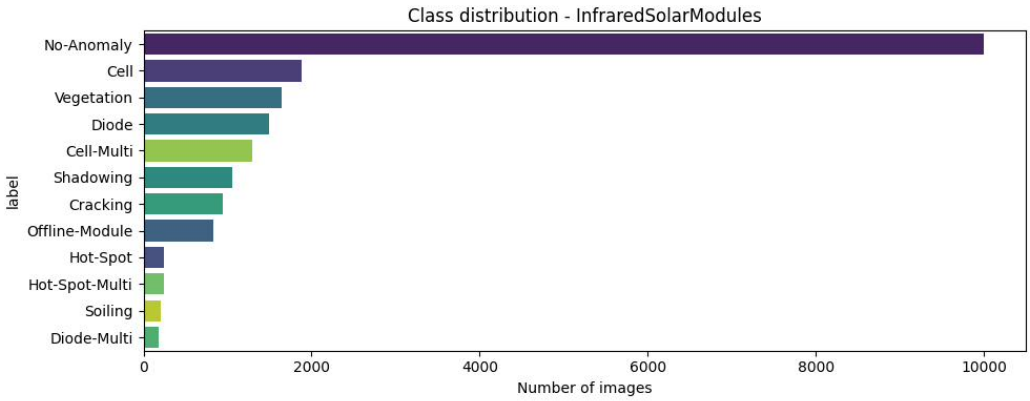

Figure 2.2: Class distribution of the InfraredSolarModules dataset (logarithmic scale).

2.4.2 SMOTE and Its Behaviour in Feature Space

The synthetic minority over-sampling technique of Chawla et al. [11] remains the most widely used data level remedy. Rather than duplicating existing minority samples, which merely reweights the loss, SMOTE generates new samples by interpolating between a minority sample and one of its k nearest same-class neighbours. For a minority sample $\mathbf{x}_i$ with a randomly chosen neighbour $\mathbf{x}_i^{\mathrm{nn}}$, a synthetic sample is

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda\big(\mathbf{x}_i^{\mathrm{nn}} - \mathbf{x}_i\big), \qquad \lambda \sim \mathcal{U}(0,1), \tag{2.8}$$

which lies on the line segment joining the two points. The procedure thereby expands the region of feature space occupied by the minority class instead of simply increasing its density at existing locations.

Elreedy and Atiya [29] provide a detailed analysis of SMOTE's behaviour and its limitations, and the wider literature is not uniformly favourable. Applied to raw pixels, interpolation between two images generally produces implausible results. More subtly, in learned embedding spaces the interpolated point is not guaranteed to be semantically valid: recent work has documented cases in which SMOTE and its variants degraded minority-class performance for deep models, because small perturbations in embedding space can correspond to large semantic shifts, leading models to overfit synthetic artefacts. Dablain et al. [30] address this by integrating the resampling into a learned encoder-decoder so that synthesis occurs in a space where interpolation is better behaved.

The implication for the present work is that applying SMOTE in the pooled CNN embedding space, as is done in Chapter 3, is a reasonable middle course, since pooled convolutional features are smoother and more semantically organised than raw pixels, but it is not automatically beneficial and must be validated empirically rather than assumed.

2.5 Feature-Level Fusion

Where several representations of the same input are available, they may be combined either at the decision level, by aggregating classifier outputs, or at the feature level, by combining representations before classification. The simplest feature level scheme is concatenation: given embeddings $\mathbf{f}_A \in \mathbb{R}^d$ and $\mathbf{f}_B \in \mathbb{R}^d$ extracted from two views of an input,

$$\mathbf{f}_{\text{fused}} = \left[\, \mathbf{f}_A \,\|\, \mathbf{f}_B \,\right] \in \mathbb{R}^{2d}, \tag{2.9}$$

after which a classifier is trained on the concatenated vector. Reviews of multimodal fusion in medical imaging note that concatenation is the most common feature level strategy and that fusing complementary representations consistently improves over any single representation, though more sophisticated interaction mechanisms can extract further gains where the representations are strongly correlated.

The premise underlying fusion is complementarity: the combination is useful when the two representations encode partially non-overlapping information, so that errors made using one may be corrected by the other. This premise is what motivates the fusion experiment in this thesis. The enhancement pipeline is not a lossless transformation. It alters the image, improving defect contrast for some classes and reducing it for others, and so the enhanced and raw representations are not redundant. A representation that performs worse than the raw image when used instead of it may nonetheless contribute information when used alongside it, and testing that possibility is a natural response to the negative substitution result. Figure 2.3 contrasts the two strategies.

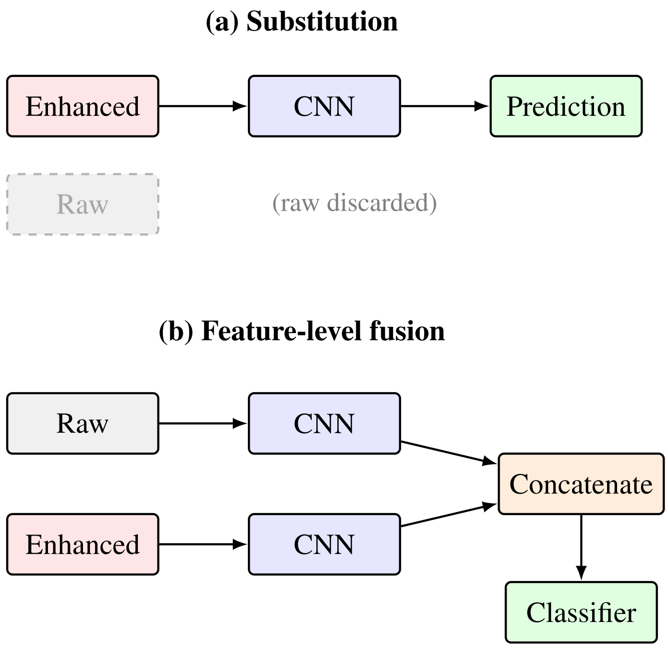

Figure 2.3: Enhanced representation used as (a) a substitute for the raw image and (b) a complementary feature through fusion.

2.6 Classical Classifiers and Ensembles over Deep Features

Once a fixed deep embedding has been produced by a frozen backbone, the final classification stage is an ordinary supervised learning problem over feature vectors, and a range of classical classifiers become applicable. This is precisely the setting of the fusion pipeline in this thesis, and the strong result of Duranay [13] on the same dataset, using deep features classified by a support vector machine, demonstrates that this two stage strategy can be highly effective. Several families of classifier are relevant.

Linear models such as logistic regression and the linear support vector machine are simple, fast, and effective on high-dimensional embeddings, though their performance depends on appropriate feature scaling. Multi-layer perceptrons add non-linear capacity through hidden layers. A separate and historically important family is the tree ensemble. Random forests [31] construct many decision trees, each trained on a bootstrap sample of the data and a random subset of features at each split, and aggregate their votes; the randomisation decorrelates the trees so that their averaged prediction has lower variance than any individual tree. Gradient-boosted trees instead build the ensemble sequentially, each new tree fitting the residual errors of the current ensemble, and the XGBoost system of Chen and Guestrin [32] made this approach both scalable and strongly regularised, which is why it and its histogram-based relatives are widely used for tabular-feature classification.

The rationale for combining several such classifiers is captured by the ensemble principle articulated by Dietterich [33]: a combination of models outperforms its members when those models are individually accurate but make different errors, since their mistakes then partly cancel under aggregation. A simple and effective realisation is soft voting, in which the predicted class probabilities of several classifiers are averaged,

$$\hat{y} = \arg\max_{c} \sum_{m=1}^{M} w_m \, p_m(c \mid \mathbf{x}), \qquad \sum_{m=1}^{M} w_m = 1, \tag{2.10}$$

where $p_m(c \mid \mathbf{x})$ is the probability assigned to class c by classifier m and w_m is its weight. This mechanism is used in Chapter 4, where classifiers of different families are combined over the fused embedding; consistent with the principle above, a boosted-tree classifier that is individually weaker than the neural classifier is nonetheless found to improve the ensemble, precisely because it errs differently.

2.7 Experimental Reliability and Statistical Validation

This section addresses a methodological concern that motivated much of the experimental design of this thesis, and which the applied PV literature treats comparatively lightly.

Training a deep network is a stochastic procedure. Weight initialisation, the order in which mini-batches are drawn, data augmentation sampling, and non-deterministic GPU kernel scheduling all introduce variability, so that two runs of an identical configuration on identical data can yield materially different models. Henderson et al. [34] demonstrated this forcefully in deep reinforcement learning, showing that reported performance differences between algorithms could be attributable to random seed selection rather than to the algorithms themselves. Subsequent work has documented the same phenomenon across supervised deep learning, and studies on small or imbalanced datasets have observed that seed variance can be large enough to reverse the ranking of competing methods.

The consequence is that a comparison between two configurations, each trained once, has limited evidential value: the observed difference confounds the effect of the intervention with the variance of the training procedure. Demšar [35] sets out the standard statistical machinery for comparing classifiers, and the appropriate instrument when two configurations are evaluated under matched conditions across several seeds is the paired t-test. Given per-seed metric values x_i for one configuration and y_i for the other, with differences $d_i = x_i - y_i$ over n seeds, the statistic is

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad \bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \qquad s_d = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(d_i - \bar{d}\right)^2}, \tag{2.11}$$

evaluated against the t distribution with $n - 1$ degrees of freedom. Pairing by seed removes the shared component of run-to-run variance and so provides more statistical power than an unpaired comparison at the same sample size.

Applying this test is uncommon in the applied PV fault detection literature, where single run comparisons remain the norm. Adopting it here is one of the methodological contributions of this thesis, and it proved decisive: as reported in Chapter 4, two single runs of the same comparison disagreed about which configuration was superior, and only the multi seed paired test resolved the question.

2.8 Explainability

A classifier that achieves high accuracy may still be relying on cues that do not correspond to the physical phenomenon of interest, such as image borders, capture artefacts, or incidental correlations in the dataset. For an inspection system intended for deployment, this possibility must be examined rather than assumed away.

Selvaraju et al. [10] introduced gradient-weighted class activation mapping (Grad-CAM), which localises the image regions responsible for a prediction without requiring architectural modification or retraining. Let A^k denote the k-th channel of the final convolutional feature map and y^c the pre-softmax score for class c. The channel importance weights are obtained by global average pooling of the gradients,

$$\alpha_k^c = \frac{1}{Z}\sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \tag{2.12}$$

where Z is the number of spatial positions, and the localisation map is the rectified weighted combination

$$L_{\text{Grad-CAM}}^c = \mathrm{ReLU}\left(\sum_k \alpha_k^c A^k\right). \tag{2.13}$$

The rectification retains only regions contributing positively to the class score. Because the method requires only a forward and backward pass through an already trained network, it is straightforward to apply to any convolutional backbone, including the EfficientNetV2-S used here.

2.9 Research Gap

Each individual component used in this thesis is, on its own, well established. Band selective wavelet enhancement rests on a clear theoretical rationale [9, 18], EfficientNetV2 is a mature and efficient backbone [1], and both feature extraction and fine tuning are standard ways to transfer a pretrained network to a new domain [26]. SMOTE is the usual remedy for class imbalance [11], feature concatenation is a common fusion strategy, tree ensembles and soft voting are dependable classifiers [31, 32, 33], and Grad-CAM is a familiar tool for explaining a model's decisions [10]. Strong results that combine deep features with a classical classifier have already been reported on the exact benchmark used here [13]. What is missing is not any single technique, but the way these techniques are evaluated and combined. Four gaps stand out, and the present work is built around closing them.

The first gap concerns evidence. Enhancement methods for thermal PV imagery are seldom tested in a way that separates a real effect from random variation. The common practice is to train a model once with the enhancement and once without it, then treat the difference as proof. As Section 2.7 shows, the run to run variation on this kind of data can be larger than the effect itself, so a single comparison of this sort proves very little. This thesis instead retrains each configuration across several random seeds and applies a paired significance test, so that the conclusion about the enhancement can actually be trusted.

The second gap concerns framing. Enhancement is nearly always judged as a replacement for the raw image, when it can equally be treated as an additional source of information. These are different questions with different answers. Showing that an enhanced image is a weaker input on its own does not show that it is useless, yet the replacement framing quietly assumes it does. Because the enhancement studied here improves contrast for some fault types and reduces it for others, the two representations are not interchangeable, and combining them is the more sensible option. This thesis tests that option directly through feature fusion.

The third gap concerns honesty about negative outcomes. A preprocessing step that fails a fair test is still informative, because it narrows what future work should try and prompts the more useful question of why it failed. Rather than quietly dropping such a result, this thesis reports it and uses it as the reason to move from substitution to fusion, which is what ultimately produces the strongest model.

The fourth gap concerns architectural diversity. Most studies on this benchmark rely on a single backbone, even though different network families tend to make different mistakes. By pairing EfficientNetV2 with ConvNeXt, two backbones with distinct inductive biases, and by reporting both the binary detection and the full multiclass diagnosis, this thesis makes fuller use of that diversity than earlier single model, single task studies on the same data.

Chapter 3

Methodology

This chapter sets out the full methodology of the thesis. It begins with a high level view of the complete pipeline and then examines each stage in turn. The order of the sections follows the flow of an image through the system: the dataset and how it is divided, the preprocessing applied to every image, the band selective wavelet enhancement, the convolutional backbones that encode the images, the transfer learning that adapts those backbones to thermal data, the correction of class imbalance, the fusion of features, the ensemble that produces the final decision, and finally the metrics, the statistical protocol, and the explainability method used to evaluate and interpret the system.

3.1 Overview of the Proposed Pipeline

The goal of the system is to take a single thermal image of a photovoltaic module and predict its condition, either as a specific fault type in the twelve class setting or as healthy against faulty in the binary setting. The design is shaped by one guiding question that runs through the whole thesis, namely whether the wavelet enhancement is more useful as a replacement for the raw image or as an additional view of it placed alongside the raw image. To answer this fairly, the pipeline keeps both the raw image and its enhanced version, encodes each with a convolutional backbone, and then studies them separately and together.

The complete flow is shown in Figure 3.1. A thermal image is first resized to a common size. It then travels along two branches. In the raw branch the image passes through unchanged, and in the enhanced branch it is processed by the band selective wavelet enhancement described in Section 3.4. Each branch is encoded by a fine tuned backbone into a pooled feature vector, and the two vectors are joined into a single fused vector. The same procedure is repeated with a second backbone of a different design, and all the resulting features are combined. The class imbalance in the training data is then corrected with SMOTE, and a weighted ensemble of classifiers produces the prediction. Each of these stages is described in detail in the sections that follow.

The pipeline is built to support the two settings studied in the thesis. In the twelve class setting the final layer distinguishes all twelve conditions, and in the binary setting the eleven fault classes are grouped together so that the task reduces to separating healthy modules from faulty ones. The binary setting answers the first and most safety critical question in field inspection, namely whether a module needs attention at all, while the twelve class setting answers the finer question of which fault is present. The same features, the same balancing, and the same ensemble structure serve both settings, with only the labels and the final decision changing between them.

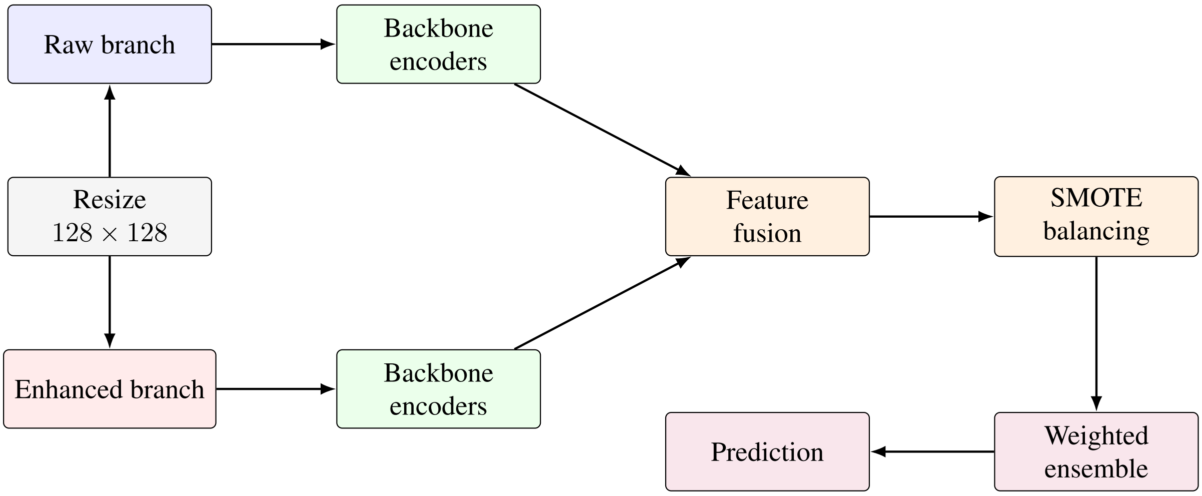

Figure 3.1: Overview of the proposed fault classification pipeline.

3.2 Dataset and Partitioning

The study uses the InfraredSolarModules dataset [8]. It contains 20,000 grayscale thermal images of individual photovoltaic modules. Each image has a native size of 24×40 pixels and is labelled into one of twelve classes, comprising eleven fault types and one healthy No Anomaly class. The dataset was released specifically to reflect the class imbalance found in real inspection data, and that imbalance is severe: the healthy class alone makes up about half of all images, while the rarest fault class holds fewer than two hundred examples.

The data is split into training, validation, and test subsets in a ratio of seventy to fifteen to fifteen. The split is stratified on the class label, so that the proportion of each class is preserved in every subset. A single fixed random seed controls the split, which means every model and every feature representation in this thesis is trained and evaluated on exactly the same partition. This is important for the fusion and ensemble stages, because they combine feature vectors sample by sample and would be corrupted if the ordering or membership of the subsets differed between runs. A check on the image file paths confirms that no image appears in more than one subset, which rules out any leakage of test data into training.

The twelve classes describe the range of conditions seen in field inspection. The No

Anomaly class covers healthy modules. The remaining eleven classes cover distinct fault types, including single and multiple cell hot spots, cracking, bypass diode faults, single and multiple diode faults, offline modules, shadowing, soiling, and vegetation intrusion. These fault types differ greatly in how often they occur and in how clearly they show up in a thermal image. Some, such as a fully offline module, produce a strong and obvious thermal signature, while others, such as light soiling, produce only a faint and diffuse pattern that is easily confused with a healthy module at this resolution. This variation in both frequency and visual clarity is the core challenge the classifier must handle, and it is the reason the later chapters place so much weight on the rare and visually subtle classes rather than on overall accuracy alone. The full class distribution is shown in Figure 3.2 on a logarithmic scale, which makes the gap between the dominant healthy class and the rarest faults clearly visible.

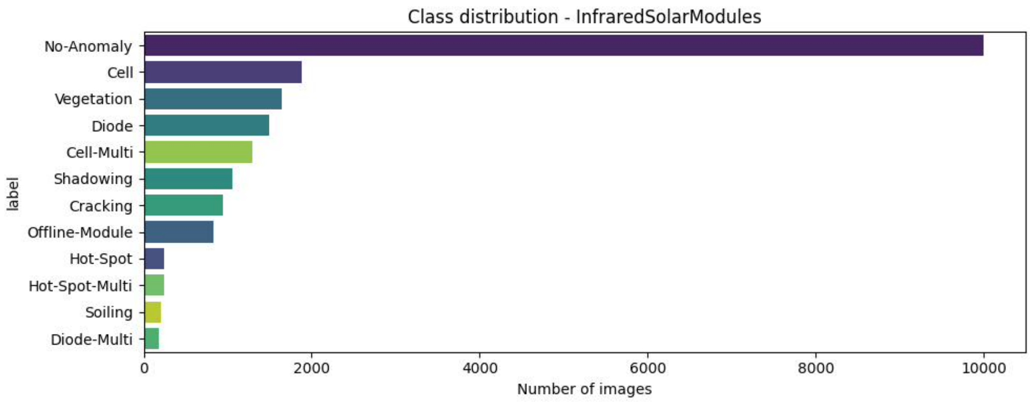

Figure 3.2: Class distribution of the InfraredSolarModules dataset (logarithmic scale).

3.3 Preprocessing

Every image is prepared in the same way before it enters either branch. The native 24×40 image is resized to 128×128 pixels using bicubic interpolation. This size is a deliberate compromise. It is large enough to allow a meaningful wavelet decomposition, since the native resolution is too small to divide into useful sub bands, and it matches the input size expected by the backbones that were pretrained on natural images. At the same time it is not so large that it invents detail that was never present in the original capture. After resizing, the pixel intensities are scaled to the range expected by each backbone. Because the backbones were trained on three channel colour images, the single grayscale channel is copied into three identical channels so that the input format matches.

3.4 Band Selective Hybrid Wavelet Enhancement

3.4.1 Why a Band Selective Design

The discrete wavelet transform separates an image into a low frequency approximation and several high frequency detail parts. This separation is useful because it lets different operations act on different kinds of image content. Contrast enhancement is helpful on the smooth, low frequency content where the overall brightness and shape of a defect live, but the same contrast enhancement applied to the whole image also amplifies the noise that sits in the high frequency detail. The design used here avoids that problem by confining contrast enhancement to the low frequency part only.

This choice was not made blindly. An earlier version of the pipeline applied contrast adjustment and sharpening one after another across the whole image, and a study of that version, reported in Section 4.1, showed that the combined cascade actually reduced image quality rather than improving it. The local contrast stretching of one stage was being magnified again by the sharpening of the next stage over the same pixels. That finding led directly to the band selective design described below, in which contrast and sharpening never act on the same frequency content.

3.4.2 Discrete Wavelet Decomposition

A single level two dimensional discrete wavelet transform is applied using the Daubechies 2 wavelet, written `db2`. The transform passes a low pass filter h and a high pass filter g along the rows and then along the columns of the image, and each filtering is followed by downsampling by a factor of two. The result is four sub bands,

$$\{\mathrm{LL},\ \mathrm{LH},\ \mathrm{HL},\ \mathrm{HH}\} = \mathrm{DWT2}(I), \tag{3.1}$$

where I is the resized image. The approximation band is produced by low pass filtering in both directions and can be written explicitly as

$$\mathrm{LL}[m,n] = \sum_{k}\sum_{l} h[k-2m]\, h[l-2n]\, I[k,l]. \tag{3.2}$$

The three detail bands follow the same pattern with the filter pairs swapped, using (h, g) for LH, (g, h) for HL, and (g, g) for HH. The LL band carries the global brightness and contrast of the module, while LH, HL, and HH carry the horizontal, vertical, and diagonal edges together with fine texture and sensor noise. Figure 3.3 shows this decomposition applied to a real thermal module image, together with the reconstructed enhanced result.

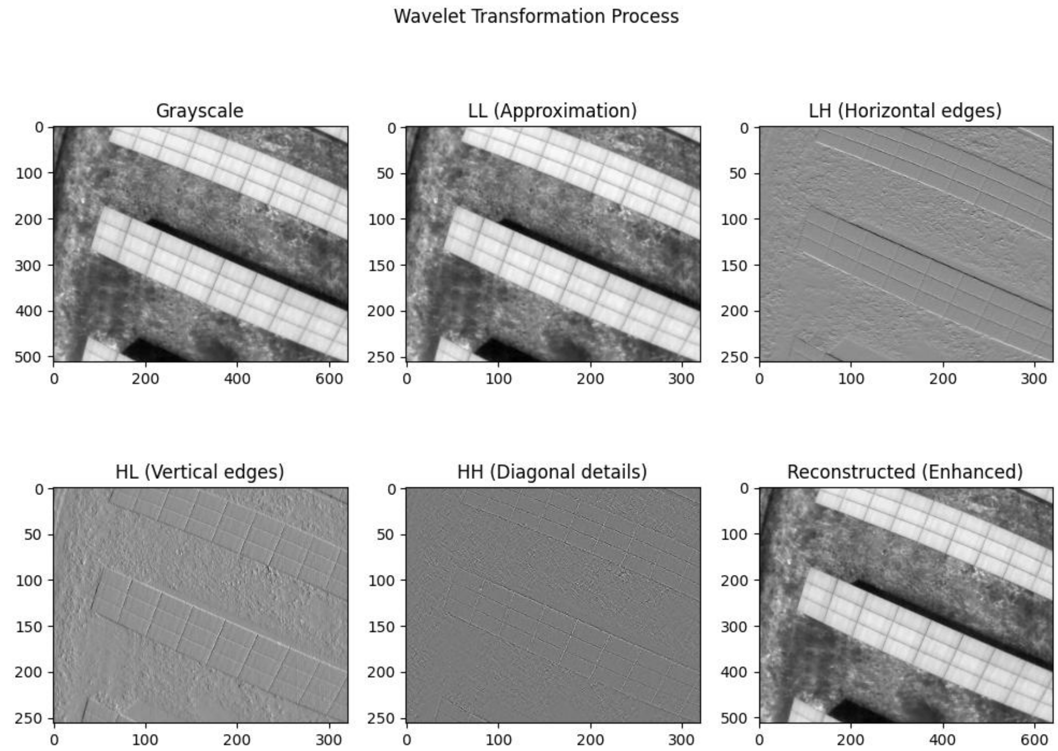

Figure 3.3: Wavelet decomposition of a thermal image into its LL, LH, HL, and HH sub bands, with the reconstructed output.

3.4.3 The Four Enhancement Stages

The enhancement has four stages, shown in Figure 3.4. The key property of the design is that the contrast operations of the second stage and the sharpening of the third stage act on separate sub bands and therefore never compound on the same pixels.

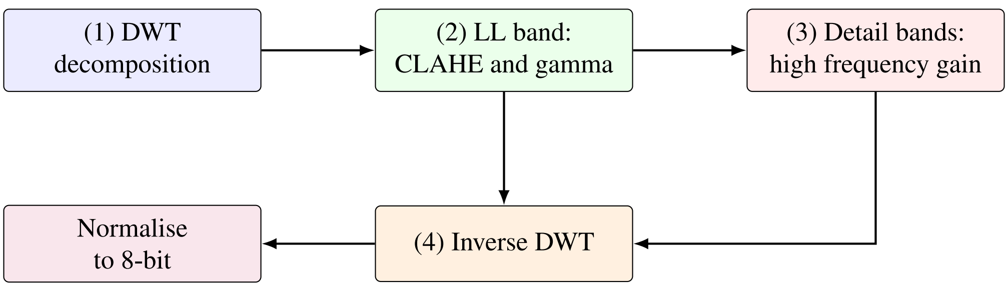

Figure 3.4: The four stage band selective enhancement.

Stage one, decomposition. The resized image is decomposed into the four sub bands of Equation (3.1).

Stage two, approximation band enhancement. Contrast limited adaptive histogram equalisation, known as CLAHE, is applied to the normalised LL band, followed by gamma correction. CLAHE improves local contrast while limiting the amplification of any single intensity, and gamma correction adjusts the overall brightness curve. The two operations combine as

$$\mathrm{LL}_{\mathrm{enh}} = \Big(\mathrm{CLAHE}_{\beta}\big(\mathrm{LL}_{\mathrm{norm}}\big)\Big)^{\gamma}, \tag{3.3}$$

with clip limit $\beta = 2.0$, a tile grid of 4×4, and gamma $\gamma = 0.85$. The result is then rescaled to the original intensity range of the LL band. Because this stage touches only the LL band, it cannot amplify the noise held in the detail bands.

Stage three, detail band enhancement. The three detail bands are multiplied by a fixed high frequency gain $g_{\mathrm{hf}} = 1.5$,

$$\mathrm{LH}_{\mathrm{enh}} = g_{\mathrm{hf}}\,\mathrm{LH}, \qquad \mathrm{HL}_{\mathrm{enh}} = g_{\mathrm{hf}}\,\mathrm{HL}, \qquad \mathrm{HH}_{\mathrm{enh}} = g_{\mathrm{hf}}\,\mathrm{HH}. \tag{3.4}$$

This sharpens edges such as cell cracks and diode boundaries. Because it acts only on the detail bands, it does not disturb the global contrast set in the previous stage.

Stage four, reconstruction. The enhanced bands are recombined by the inverse transform,

$$I_{\mathrm{enh}} = \mathrm{IDWT2}\big(\mathrm{LL}_{\mathrm{enh}},\ \mathrm{LH}_{\mathrm{enh}},\ \mathrm{HL}_{\mathrm{enh}},\ \mathrm{HH}_{\mathrm{enh}}\big), \tag{3.5}$$

and the reconstructed image is normalised to the eight bit range and copied across three channels, ready for the backbone. Figure 3.5 shows the visual effect of the enhancement, with one example from each of the twelve classes placed next to its raw version.

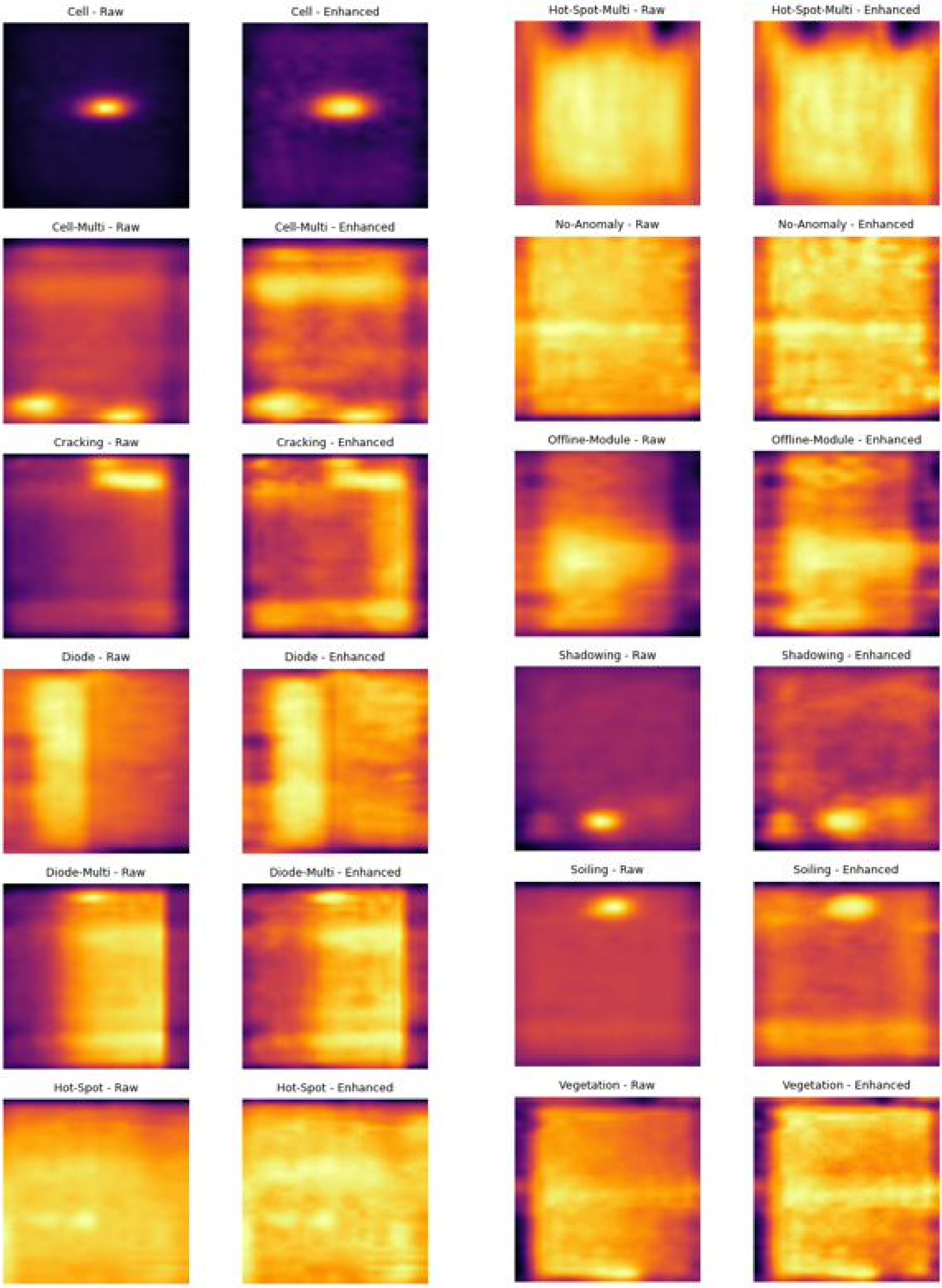

Figure 3.5: Raw and enhanced thermal images, one example per class.

3.4.4 Enhancement Quality Metrics

To measure what the enhancement does to the image, three standard quality metrics are used per class. The contrast to noise ratio compares the intensity difference between a defect region and its background against the background variation,

$$\mathrm{CNR} = \frac{|\mu_{\mathrm{fg}} - \mu_{\mathrm{bg}}|}{\sigma_{\mathrm{bg}}}, \tag{3.6}$$

where μ_{fg} and μ_{bg} are the mean intensities of the foreground and background and σ_{bg} is the background standard deviation. The structural similarity index compares the enhanced image with the original in terms of luminance, contrast, and structure, and the peak signal to noise ratio measures the fidelity of the enhanced image to the original. These metrics describe the enhancement rather than prove it is best. As Chapter 4 shows, the enhancement raises contrast for some classes and lowers it for others, and this uneven behaviour is part of the reason the fusion approach is preferred over simple substitution.

3.5 Convolutional Neural Networks and Backbones

3.5.1 Convolutional Neural Network Fundamentals

A convolutional neural network learns useful features directly from image pixels rather than relying on features designed by hand. Its central operation is convolution, in which a small learnable kernel slides across the input and computes a weighted sum at each position. For an input feature map X and a kernel W, the output at position (i, j) in output channel o is

$$Y_o[i, j] = \sigma\left(b_o + \sum_c \sum_u \sum_v W_{o,c}[u, v]\, X_c[i + u,\, j + v] \right), \tag{3.7}$$

where c runs over input channels, b_o is a bias term, and σ is a nonlinear activation. A common choice of activation is the rectified linear unit,

$$\sigma(z) = \max(0, z), \tag{3.8}$$

which keeps positive values and sets negative ones to zero. Stacking many convolutional layers lets the network build a hierarchy of features, from simple edges in the early layers to complex, class specific shapes deeper in the network. Pooling layers reduce the spatial size of the feature maps and give the network a degree of tolerance to small shifts in the input. After the final convolutional stage, a global pooling step averages each feature map into a single number, producing the fixed length feature vector that represents the image. It is this vector, rather than the raw image, that the classifiers in this thesis work with. Figure 3.6 shows this flow in a simplified form, from the input image through alternating convolution and pooling stages to the pooled feature vector and the final class scores.

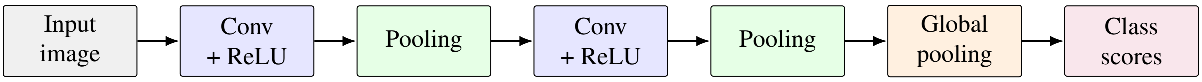

Figure 3.6: Simplified flow of a convolutional neural network.

3.5.2 EfficientNetV2-S

The first backbone is EfficientNetV2-S [1]. The EfficientNet family introduced compound scaling, in which the depth, the width, and the input resolution of the network are grown together by a single coefficient rather than adjusted one at a time [24]. EfficientNetV2 refined this idea using a training aware architecture search and by placing fused mobile inverted bottleneck blocks, called Fused-MBConv, in the early stages of the network, which train faster on modern hardware than the standard MBConv blocks used later. The overall structure is shown in Figure 3.7. The network begins with a small stem convolution, passes through a sequence of Fused-MBConv and MBConv stages that gradually reduce spatial size while increasing the number of channels, and ends with a final convolution, global pooling, and a classification layer.

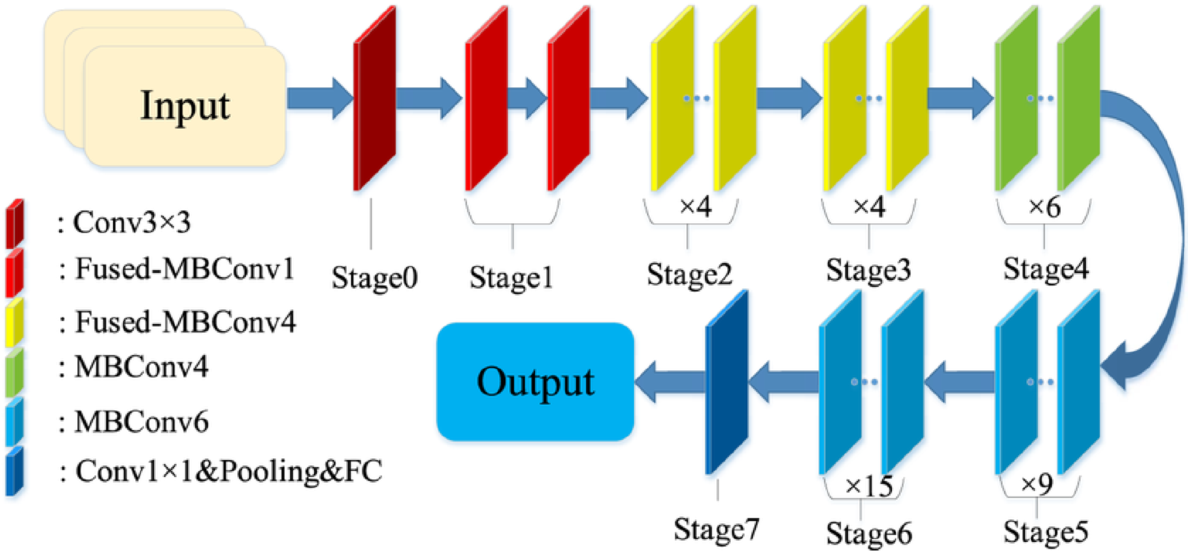

Figure 3.7: Structure of the EfficientNetV2-S backbone [1].

A useful part of these blocks is the squeeze and excitation mechanism, which lets the network reweight its channels according to their importance. It first averages each channel over space, then learns a small gating function that scales each channel,

$$s_c = \sigma\big(W_2\,\delta(W_1\,z_c)\big), \qquad \tilde{X}_c = s_c\,X_c, \tag{3.9}$$

where z_c is the spatial average of channel c, δ is a rectified linear unit, σ is a sigmoid, and W_1 and W_2 are learnable weights. This helps the network emphasise the channels that matter for a given image.

The two block types that make up the network differ in how they treat the early expansion of channels. A standard MBConv block first expands the channels with a pointwise convolution, then applies a depthwise convolution, a squeeze and excitation step, and a final pointwise convolution that projects the channels back down, with a residual connection across the block. A Fused-MBConv block replaces the first pointwise convolution and the depthwise convolution with a single ordinary convolution, which is less efficient in theory but faster in practice on modern accelerators for the small feature maps found in the early stages. EfficientNetV2 uses Fused-MBConv blocks in its early stages and standard MBConv blocks in its later stages, which is one of the design choices that lets it train quickly while keeping a small parameter count. In this thesis the network is used both as a fixed feature extractor and, in the final configuration, as a backbone that is fine tuned on the thermal images.

3.5.3 ConvNeXt-Tiny

The second backbone is ConvNeXt-Tiny, a convolutional network that borrows several ideas from vision transformers while remaining fully convolutional. It is organised into four stages with a block distribution of three, three, nine, and three, and channel widths of ninety six, one hundred ninety two, three hundred eighty four, and seven hundred sixty eight. A patchify stem, a four by four convolution with stride four, first divides the image into non overlapping patches. Each ConvNeXt block then applies a large seven by seven depthwise convolution, a layer normalisation, a pointwise convolution that expands the channels by a factor of four, a GELU activation, and a second pointwise convolution that projects the channels back, with a residual connection adding the input of the block to its output. Between stages, a downsampling layer halves the spatial size and doubles the channels. This block structure is drawn in Figure 3.8.

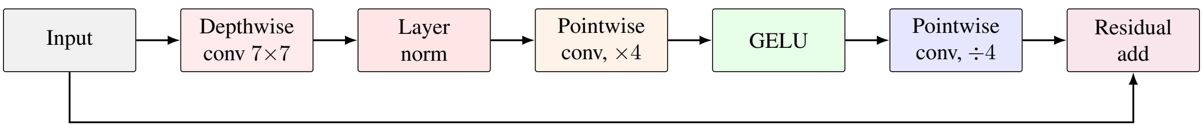

Figure 3.8: Structure of a ConvNeXt block with its residual connection.

Because the two backbones are built on different principles, EfficientNetV2 on compound scaled bottleneck blocks and ConvNeXt on large kernel depthwise blocks, they tend to make different mistakes. That difference is exactly what makes them work well together in the ensemble of Section 3.9.

3.6 Transfer Learning

Both backbones are started from weights pretrained on the ImageNet dataset [25] and then adapted to thermal images. Transfer learning is appropriate here because the early features learned by a convolutional network, such as edges and simple textures, are general and carry over well from natural images to thermal images [26], which lets a large network be trained effectively on a relatively small thermal dataset. Two regimes of transfer learning are used at different points in the study, and they are shown in Figure 3.9.

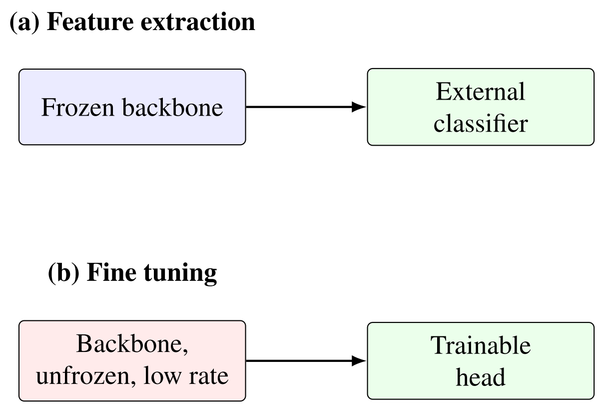

Figure 3.9: The two transfer learning regimes used in the study.

In the feature extraction regime the backbone is frozen and used only to turn each image into a fixed feature vector, and separate classifiers are trained on top of those vectors. In the fine tuning regime the backbone is unfrozen and trained end to end on the thermal images at a low learning rate, so that its internal features adjust to the thermal domain instead of staying tuned to natural images. Fine tuning uses the Adam optimiser, a class weighted cross entropy loss to counter the imbalance, light data augmentation in the form of horizontal flips and small rotations, and early stopping on the validation accuracy with the best checkpoint kept. The class weighted cross entropy loss for one sample with true class y and predicted probabilities $\hat{p}$ is

$$\mathcal{L} = -\sum_{c=1}^{C} w_c \mathbb{1}[y = c] \, \log \hat{p}_c, \tag{3.10}$$

where w_c is a weight inversely related to the frequency of class c, so that rare classes contribute more to the loss. The final results use features from the fine tuned backbones, since adapting the features to the thermal domain was found to improve performance clearly over frozen features.

3.7 Class Imbalance Correction

The training set inherits the strong imbalance of the dataset, and a classifier trained on it without correction tends to favour the healthy class and neglect the rare faults. To counter this, the synthetic minority over sampling technique, SMOTE [11], is applied in the fused feature space. Rather than copying existing minority samples, SMOTE creates new ones by interpolating between a minority sample and one of its nearest neighbours of the same class. For a minority sample $\mathbf{x}_i$ and a randomly chosen same class neighbour $\mathbf{x}_i^{\mathrm{nn}}$, a synthetic sample is

$$\mathbf{x}_{\mathrm{new}} = \mathbf{x}_i + \lambda \left(\mathbf{x}_i^{\mathrm{nn}} - \mathbf{x}_i\right), \qquad \lambda \sim \mathcal{U}(0,1). \tag{3.11}$$

Generating samples in the feature space rather than in pixel space avoids producing unrealistic images while still balancing the training distribution. Each minority class is raised to a fixed ceiling rather than all the way to the size of the majority class, which balances the classes while limiting the number of purely synthetic samples. Importantly, oversampling is applied only to the training partition. The validation and test partitions keep their natural distribution, so that the reported performance reflects the real class proportions.

3.8 Feature Fusion and Classification

For each image, the raw branch and the enhanced branch are each turned into a pooled feature vector by a backbone. The two vectors are joined end to end by concatenation,

$$\mathbf{f}_{\mathrm{fused}} = \left[\, \mathbf{f}_{\mathrm{raw}} \parallel \mathbf{f}_{\mathrm{enh}} \,\right], \tag{3.12}$$

so that the fused vector holds both the original view and the enhanced view of the same image. The fused vectors from both backbones are then concatenated in the same way, so that the final representation combines two image views and two network architectures. Before classification the fused features are standardised to zero mean and unit variance,

$$\tilde{x}_j = \frac{x_j - \mu_j}{s_j}, \tag{3.13}$$

where μ_j and s_j are the mean and standard deviation of feature j estimated on the training set only. Standardisation matters because the classifiers that follow are sensitive to the scale of their inputs. Three classifiers are trained on the balanced, standardised features: a multilayer perceptron with two hidden layers, a logistic regression model, and a histogram based gradient boosted tree ensemble [32]. These belong to three different families, neural, linear, and tree based, and are chosen so that their errors are as different as possible, which is what makes combining them worthwhile.

3.9 Multi-Backbone Ensemble

The final prediction is made by combining the three classifiers through weighted soft voting. Each classifier outputs a probability for every class, and these probabilities are averaged with weights,

$$\hat{y} = \arg\max_c \sum_{m=1}^{M} w_m \, p_m(c \mid \mathbf{x}), \qquad \sum_{m=1}^{M} w_m = 1, \tag{3.14}$$

where $p_m(c \mid \mathbf{x})$ is the probability that classifier m assigns to class c, and w_m is its weight. The weights are chosen by a grid search on the validation set to maximise the macro F1 score, and the chosen weights are then applied once to the test set. Keeping the search on the validation data ensures that the reported test performance is not tuned to the test set. Soft voting is preferred over hard voting because averaging the probabilities uses the confidence of each classifier rather than only its top choice, which usually gives a more stable result.

3.10 Binary Fault Detection

Alongside the twelve class diagnosis, the system is also evaluated as a binary detector that separates healthy modules from faulty ones. This follows the two level view set out in Chapter 2, where binary detection answers the first question an operator asks, namely whether a module is faulty at all, and the multiclass diagnosis answers the harder question of which fault is present. The binary task is built on exactly the same features and the same ensemble as the twelve class task. The only change is in the labels: the No Anomaly class becomes the healthy class, and the eleven fault types are merged into a single faulty class. The classifier is then trained to separate these two groups. Because collapsing the fault types removes the difficult confusions between faults that look alike, the binary task is easier than the twelve class task, and it is expected to reach a higher score. For this task the area under the receiver operating characteristic curve is reported together with accuracy, precision, recall, and F1, and particular attention is paid to the recall on the faulty class, since a missed fault is more costly in practice than a false alarm.

3.11 Evaluation Metrics

Performance is reported with overall accuracy and with macro averaged precision, recall, and F1 score. For a class c, precision measures how many of the samples predicted as c truly belong to c, recall measures how many of the true c samples are found, and the F1 score is their harmonic mean,

$$P_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c}, \qquad R_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}, \qquad \mathrm{F1}_c = \frac{2\, P_c\, R_c}{P_c + R_c}, \tag{3.15}$$

where TP, FP, and FN are the counts of true positives, false positives, and false negatives. The macro F1 score is the plain average of the per class F1 scores,

$$\mathrm{F1}_{\mathrm{macro}} = \frac{1}{C}\sum_{c=1}^{C}\mathrm{F1}_c, \tag{3.16}$$

with $C = 12$. Macro averaging gives every class the same weight regardless of its size, so poor performance on a rare fault is not hidden by strong performance on the large healthy class. For this reason macro F1 is treated as the primary metric throughout the thesis. In the binary setting, where the task is only to separate healthy from faulty, the area under the receiver operating characteristic curve is also reported, since it summarises the trade off between true and false positive rates across all decision thresholds.

3.12 Statistical Validation Protocol

Training a deep network on a small and imbalanced dataset is a stochastic process, and a single run can give a misleading impression. To guard against this, the comparison between the enhanced and raw configurations is not based on one run. Each configuration is trained independently at three random seeds, and the resulting scores are compared with a paired t-test across the seeds. Pairing by seed removes the part of the variation that the two configurations share, which gives the test more power than an unpaired comparison. The paired t statistic is

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \tag{3.17}$$

where $\bar{d}$ is the mean of the per seed differences between the two configurations, s_d is their standard deviation, and n is the number of seeds. This protocol is what allows the thesis to state whether a difference between the enhanced and raw settings is a real effect or simply the result of random variation in training.

3.13 Explainability

To check that the trained network makes its decisions for sensible reasons, the gradient weighted class activation mapping method, Grad-CAM [10], is applied to the fine tuned backbone. Grad-CAM produces a heatmap that shows which parts of an image most influenced the predicted class. It first computes the gradient of the class score with respect to the final convolutional feature map and averages these gradients over space to obtain an importance weight for each channel,

$$\alpha_k^c = \frac{1}{Z}\sum_i\sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \tag{3.18}$$

where A^k is the k-th channel of the feature map, y^c is the score for class c, and Z is the number of spatial positions. The heatmap is then the rectified weighted sum of the feature map channels,

$$L^c_{\text{Grad-CAM}} = \mathrm{ReLU}\left(\sum_k \alpha^c_k\, A^k\right). \tag{3.19}$$

The rectifier keeps only the regions that push the score up for the class of interest. Overlaying this heatmap on the input image shows whether the network is responding to the genuine defect region or to something irrelevant such as the module frame, which gives a qualitative check to accompany the numerical metrics.

3.14 Implementation Environment

The system was implemented in Python. The wavelet decomposition used the PyWavelets library, and the CLAHE and image operations used OpenCV. The EfficientNetV2-S backbone was implemented in TensorFlow and Keras, while the ConvNeXt-Tiny backbone used PyTorch together with the `timm` library. SMOTE was provided by the imbalanced-learn library, and the classifiers, feature scaling, and metrics used scikit-learn. Training was carried out on cloud graphics processing unit runtimes. To make the experiments practical, the feature vectors from each backbone were computed once and cached to disk, so that the fusion, balancing, and ensemble stages could be repeated quickly without re-running the costly backbone inference each time.

Chapter 4

Results and Discussion

This chapter presents the results in the order the work developed. It begins with the enhancement, showing through an ablation study how the design was chosen and how it affects the thermal images. It then reports the binary detection of healthy against faulty modules. After that it compares the raw and enhanced inputs directly, and then follows, step by step, how the accuracy was raised from the simple baselines to the final model, with the confusion matrix shown at every stage. It closes with the per class analysis, an explainability check, and the final confusion matrix.

4.1 The Enhancement and its Ablation

The first question was how to enhance the thermal images, and whether the enhancement helps at all. The design was chosen through an ablation study that measured image quality with three standard measures: the contrast to noise ratio, which reflects how clearly a defect stands out from its background, the structural similarity index, and the peak signal to noise ratio, which both reflect how close the enhanced image stays to the original. Figure 4.1 compares the raw image against basic wavelet, adaptive wavelet, homomorphic, and unsharp mask enhancement.

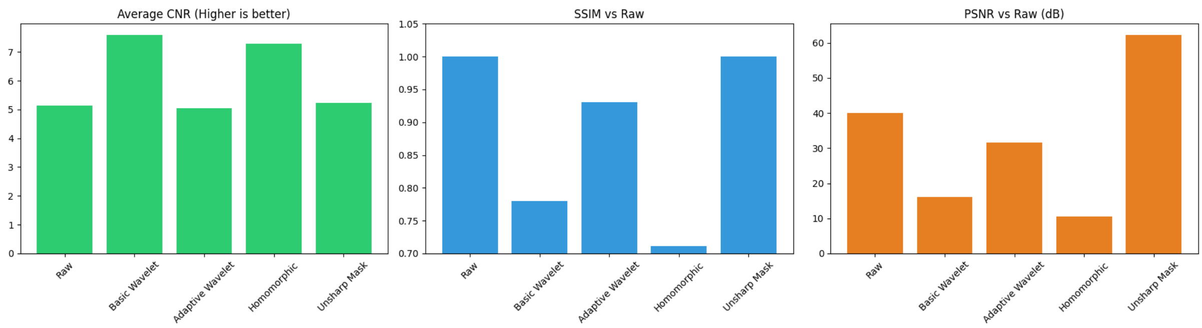

Figure 4.1: Ablation of enhancement methods on CNR, SSIM, and PSNR.

The three measures do not agree, and that disagreement is the key point. The wavelet methods give the best contrast to noise ratio, which is the measure that matters for spotting defects, but because they change the image more, they score lower on similarity and signal to noise. Methods that barely touch the image, such as unsharp mask, keep a high similarity but add little contrast. This is why image quality alone cannot tell us whether an enhancement helps the classifier, and it is why an earlier design that stacked several operations one after another was dropped. That cascade magnified its own contrast changes and added artefacts, so the final design keeps the contrast work and the sharpening on separate frequency bands.

Figure 4.2 looks closer at the chosen enhancement across all twelve classes, comparing the contrast gain method by method. It shows that the enhancement raises contrast for some classes but lowers it for others. This uneven effect is important, because it means the raw and enhanced images are not the same information, and this is the observation the later fusion step is built on.

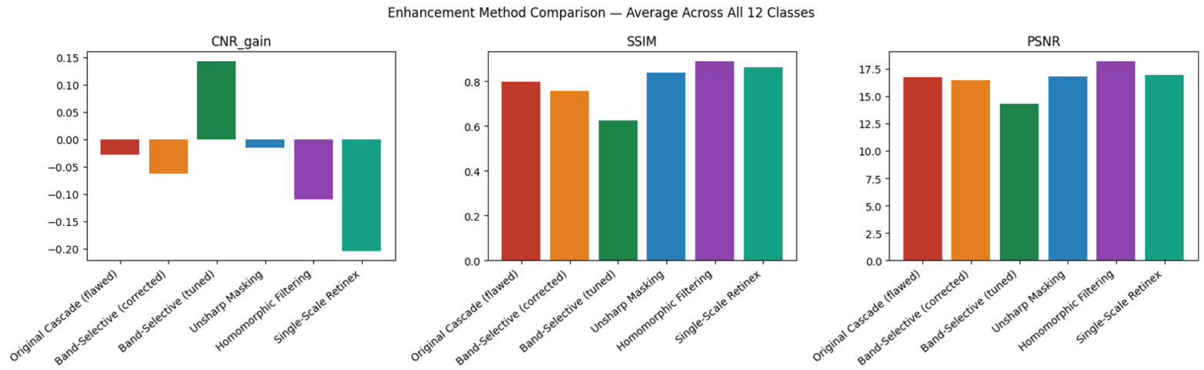

Figure 4.2: Enhancement effect across methods, averaged over the twelve classes.

4.2 Binary Fault Detection

The first classification task, and the most important one in the field, is to tell a healthy module from a faulty one. The eleven fault classes were merged into a single faulty class, and the classifier was trained to separate healthy from faulty using the fused features of the full system.

The binary detector worked very well. It reached an accuracy of 0.930 and an area under the receiver operating characteristic curve of 0.977. Figure 4.3 shows the confusion matrix. Of the 1500 healthy modules, 1414 were found correctly, and of the 1500 faulty modules, 1375 were found correctly. The recall on the faulty class was 91.7 percent, so the system catches most real faults. This is the number that matters most in practice, since missing a fault is worse than a false alarm.

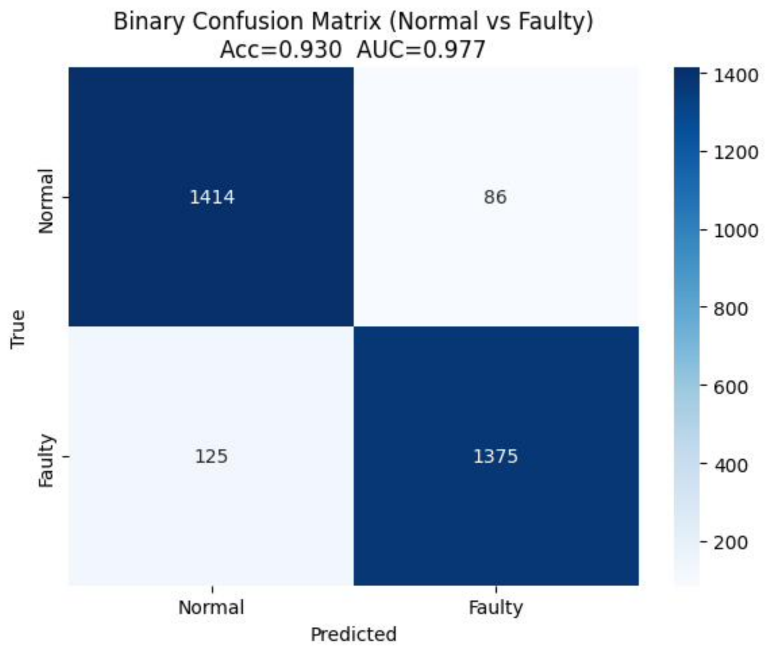

Figure 4.3: Binary confusion matrix for healthy against faulty detection.

4.3 Raw against Enhanced

The next question was whether the enhancement, used in place of the raw image, helps the twelve class classifier. Figure 4.4 compares the two directly on accuracy, precision, recall, and F1, and their confusion matrices are shown in Figure 4.5 and Figure 4.6.

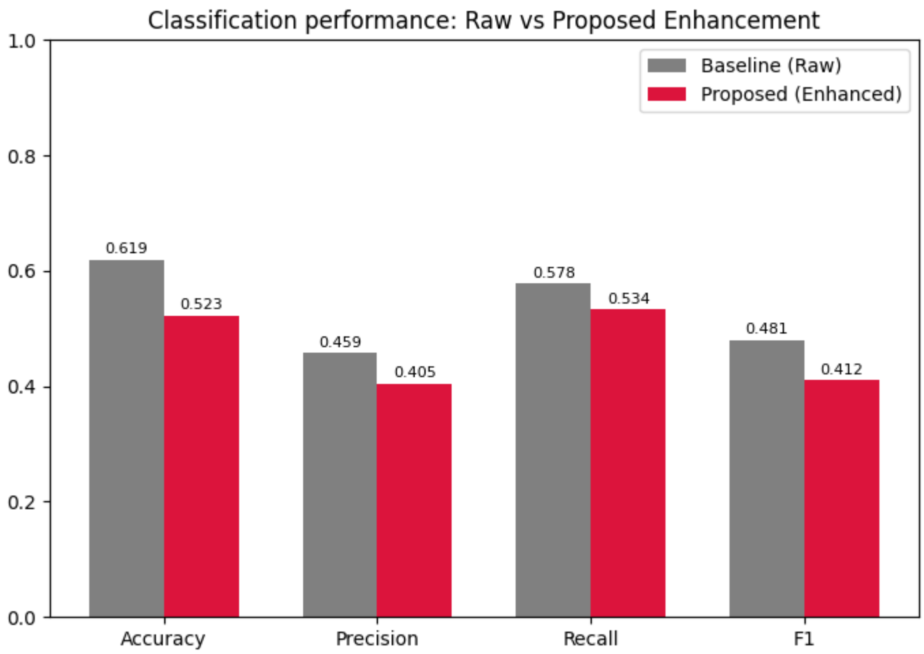

Figure 4.4: Classification performance of the raw and enhanced inputs.

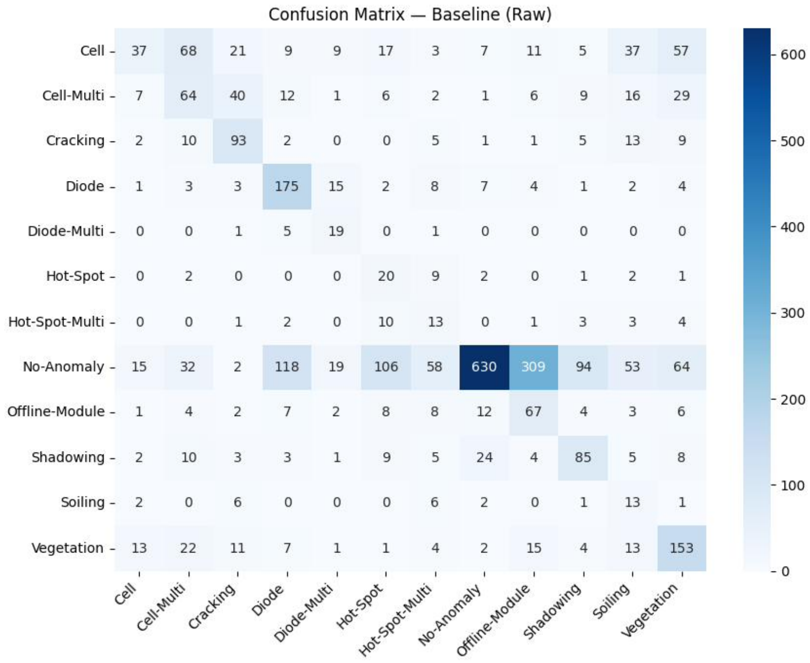

Figure 4.5: Confusion matrix for the raw input, at 0.619 accuracy.

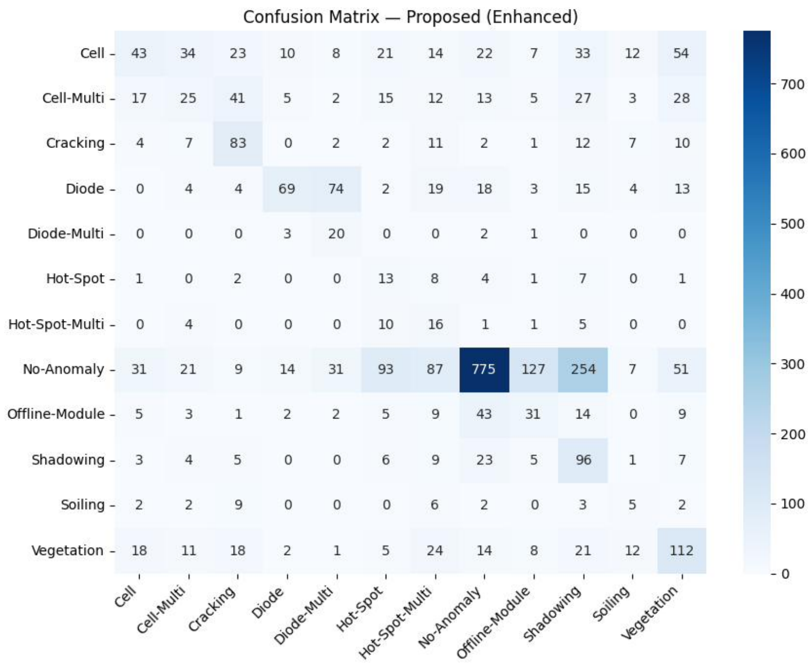

Figure 4.6: Confusion matrix for the enhanced input, at 0.523 accuracy.

The enhanced input is not clearly better than the raw input, and both matrices lean towards the large healthy class and struggle with the rare faults. A single run could not settle the comparison, because training on this small dataset is noisy. Two runs of the same enhanced setup gave accuracies of 53.6 and 42.9 percent, a gap wide enough to flip the conclusion. So both setups were trained across three seeds and compared with a paired test. The accuracy difference was not significant, with a p value of 0.461, while the raw input was significantly better on macro F1, with a p value of 0.028. The enhancement does not help as a substitute for the raw image. That negative result is what pushed the work towards fusion.

4.4 Building Up the Accuracy

Because the enhancement changes the image unevenly, the raw and enhanced views hold different information and can be joined rather than chosen between. From this idea the accuracy was raised in clear steps. Table 4.1 lists every stage, and each stage is explained below with its confusion matrix.

Table 4.1: Accuracy and macro F1 at each stage of the method.

Method	**Accuracy**	**Macro F1**
Enhanced (wavelet) alone	0.523	0.412
Raw alone	0.619	0.481
Fusion (frozen features)	0.694	0.528
Fusion + SMOTE + MLP	0.738	0.551
Frozen features + ensemble	0.768	0.574
Fine tuned EfficientNetV2 + fusion	0.784	0.629
EfficientNetV2 + ConvNeXt ensemble	0.808	0.662
+ weighted vote (final)	**0.811**	**0.665**

4.4.1 Fusion of Raw and Enhanced

The first step fused the raw and enhanced features instead of using either alone. On their own the enhanced input reached 0.523 and the raw input 0.619, but fusing the two raised the accuracy to 0.694. This jump is the clearest proof of the main idea, that the enhancement fails as a replacement but helps as a complement. Figure 4.7 shows the confusion matrix, where the diagonal is already stronger than for either single input.

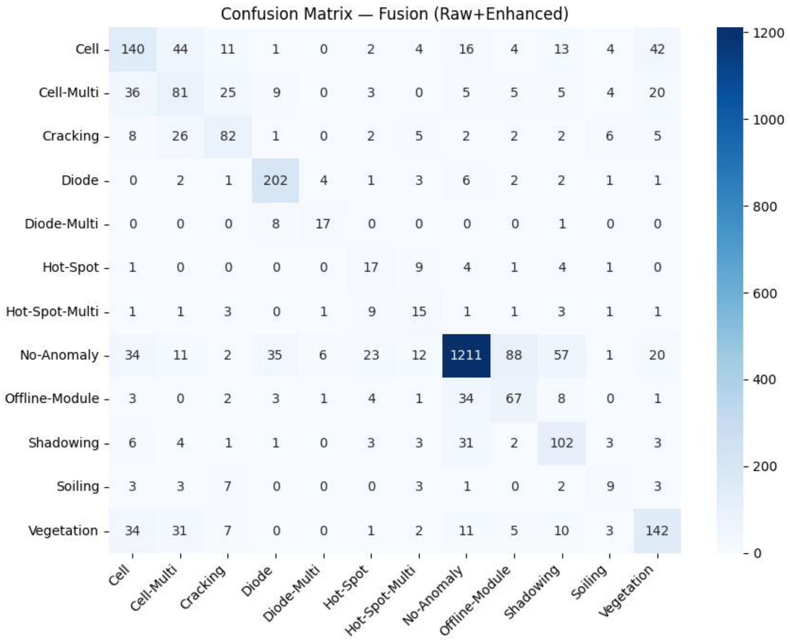

Figure 4.7: Confusion matrix for the fused raw and enhanced features, at 0.694 accuracy.

4.4.2 Balancing with SMOTE

The next step handled the class imbalance. SMOTE created synthetic samples for the rare classes in the feature space, and a multilayer perceptron was trained on the balanced features. This raised the accuracy to 0.738 and improved the macro F1, which rewards better results on the rare classes. Figure 4.8 shows that several smaller classes are recognised better than before.

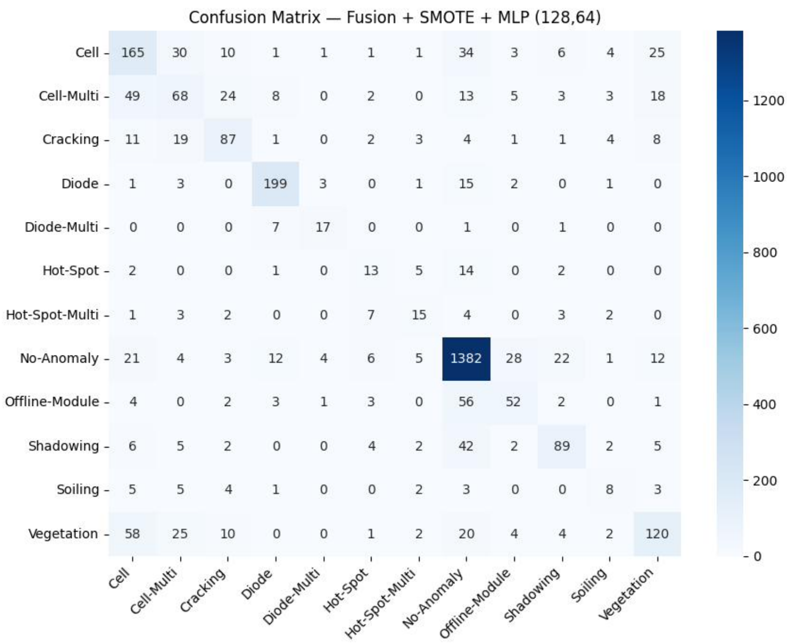

Figure 4.8: Confusion matrix after fusion, SMOTE, and a multilayer perceptron, at 0.738 accuracy.

4.4.3 Ensemble of Classifiers

Combining several classifiers of different families into an ensemble raised the accuracy to 0.768 and the macro F1 to 0.574. The ensemble is steadier than any single classifier because the members make different mistakes, so their errors partly cancel. Figure 4.9 shows the confusion matrix at this stage.

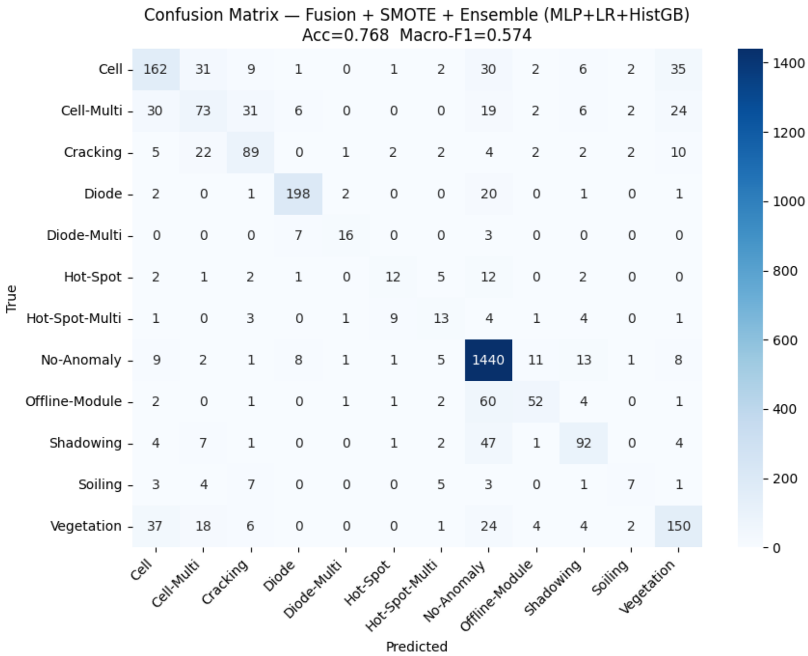

Figure 4.9: Confusion matrix for the frozen feature ensemble, at 0.768 accuracy.

4.4.4 Fine Tuning the Backbone

The largest gain in macro F1 came from fine tuning the backbone on the thermal images, rather than using features from a network trained only on natural images. This adapted the features to the thermal domain and lifted the result to 0.784 accuracy and 0.629 macro F1. Figure 4.10 shows the clearer diagonal that follows.

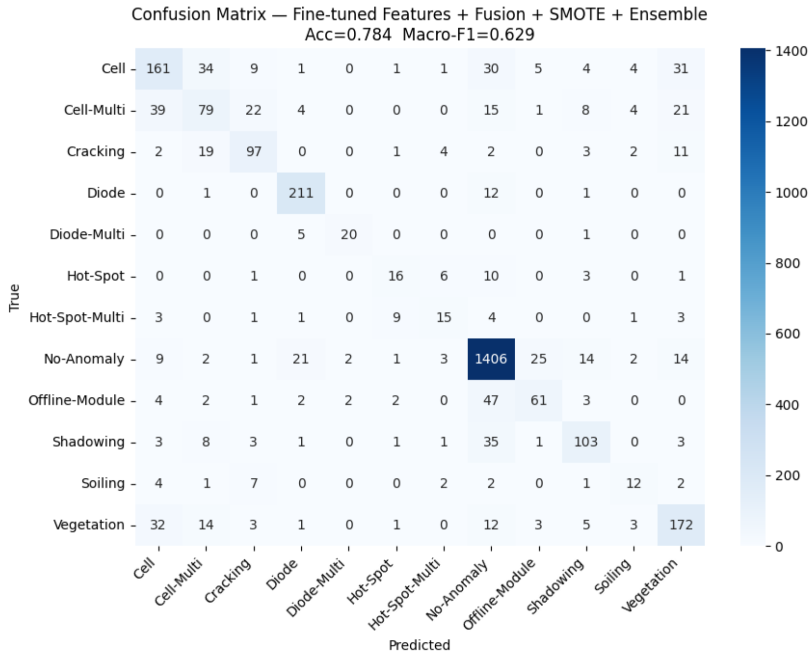

Figure 4.10: Confusion matrix after fine tuning the backbone, at 0.784 accuracy.

4.4.5 Two Backbones

Adding a second backbone of a different design, ConvNeXt, alongside EfficientNetV2 helped, because the two networks make different mistakes and their features complement each other. This raised the accuracy to 0.808 and the macro F1 to 0.662. Figure 4.11 shows the confusion matrix for the two backbone ensemble.

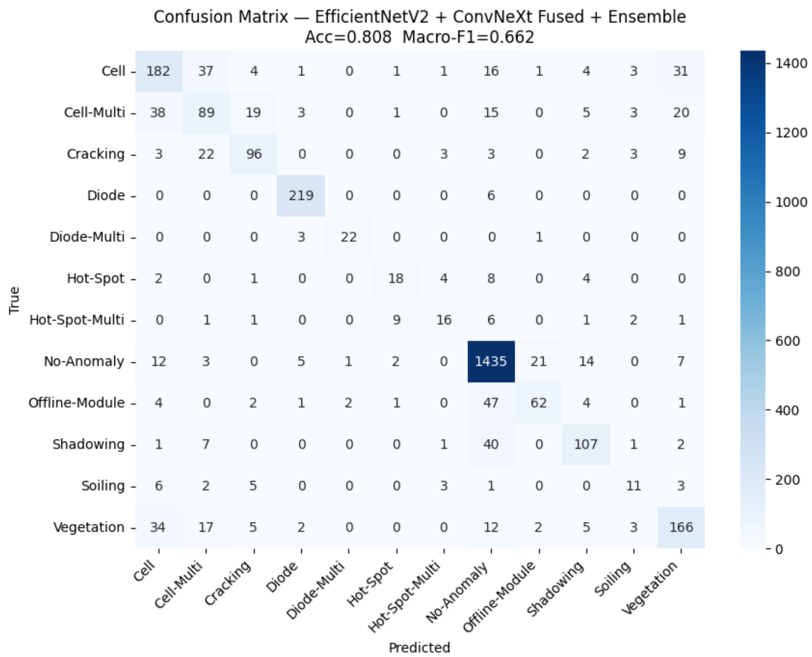

Figure 4.11: Confusion matrix for the two backbone ensemble, at 0.808 accuracy.

4.4.6 Weighted Voting and the Final Model

The last step chose the vote weights of the ensemble on the validation set, which gave a small further gain to the final result of 0.811 accuracy and 0.665 macro F1. Figure 4.12 shows the whole climb from 0.523 to 0.811, and the final confusion matrix is shown later in Figure 4.17.

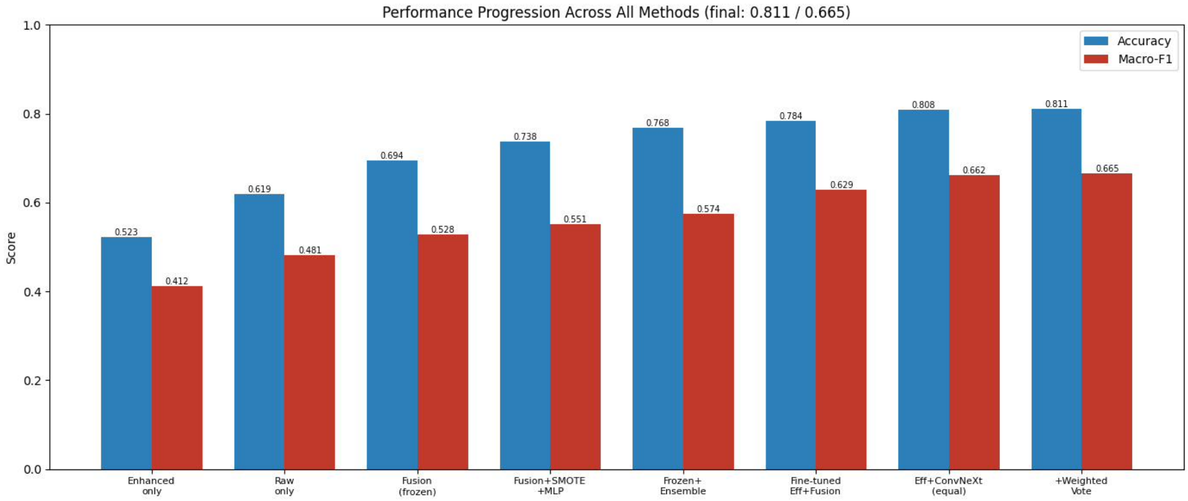

Figure 4.12: Accuracy and macro F1 across all stages of the method.

Figure 4.13 shows the training and validation curves of the fine tuned backbone. The two curves rise together and stay close, so the model learns steadily and does not overfit badly on the small dataset.

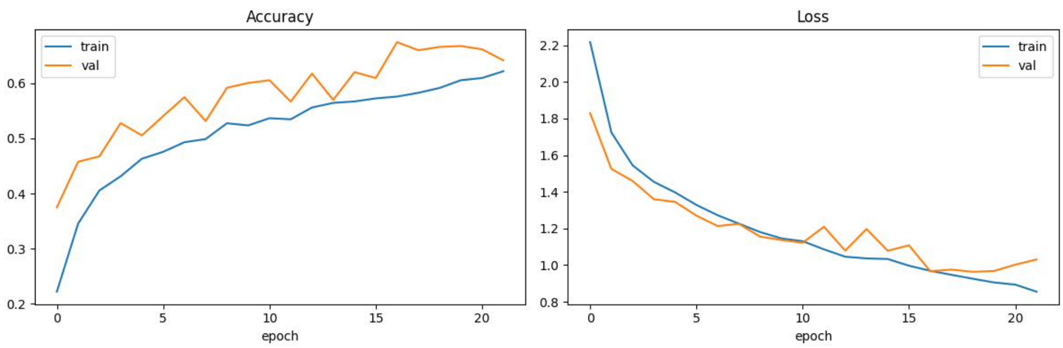

Figure 4.13: Training and validation accuracy and loss for the fine tuned backbone.

4.5 Per Class Results

Because the data is imbalanced, the per class scores matter as much as the overall accuracy. Figure 4.14 compares the single backbone model with the final two backbone ensemble, from the rarest class to the most common.

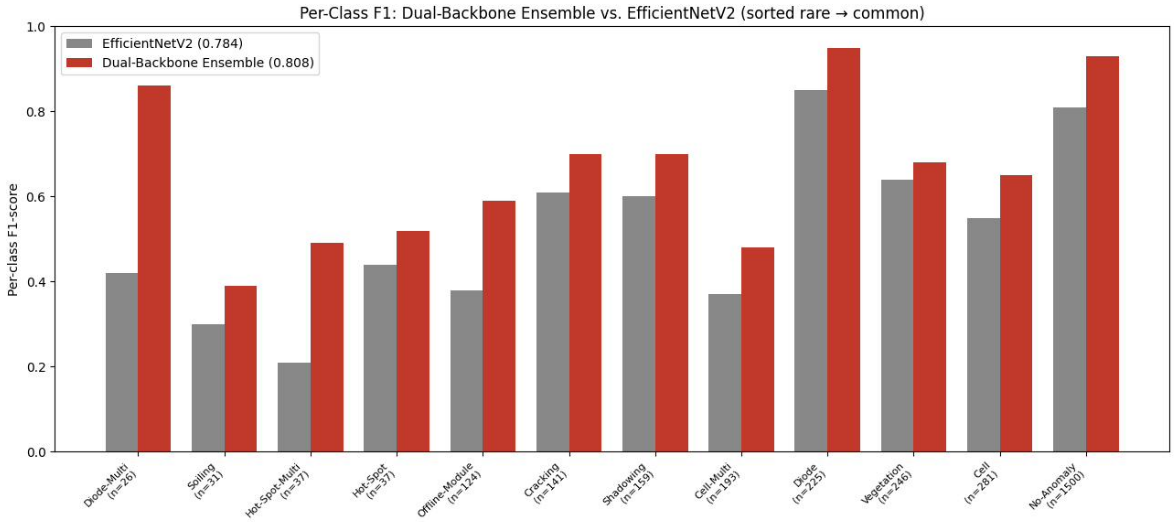

Figure 4.14: Per class F1 for the single backbone and the final two backbone ensemble.

The two backbone ensemble improves most classes, and the gains are clearest on the rare faults. Diode Multi, with only twenty six test samples, reaches an F1 of 0.92, and Cracking, Shadowing, and Offline Module also improve. The common classes stay strong, with Diode above 0.95 and No Anomaly at 0.93. The hardest class is Soiling, which stays near 0.43 because it has few samples and a faint, spread out signature. This shows that the remaining difficulty comes from the data itself, from the low resolution and the rare classes, and not from the choice of model.

4.6 Explainability

To check that the model looks at the right place, Grad-CAM heatmaps were made for the fine tuned backbone. Figure 4.15 shows some examples next to the original images. For the correctly classified faults, the bright region sits on the real defect, such as the hot cell in a hot spot image or the affected area in a shadowing case, and not on the frame or background. This gives confidence that the model responds to real thermal signs.

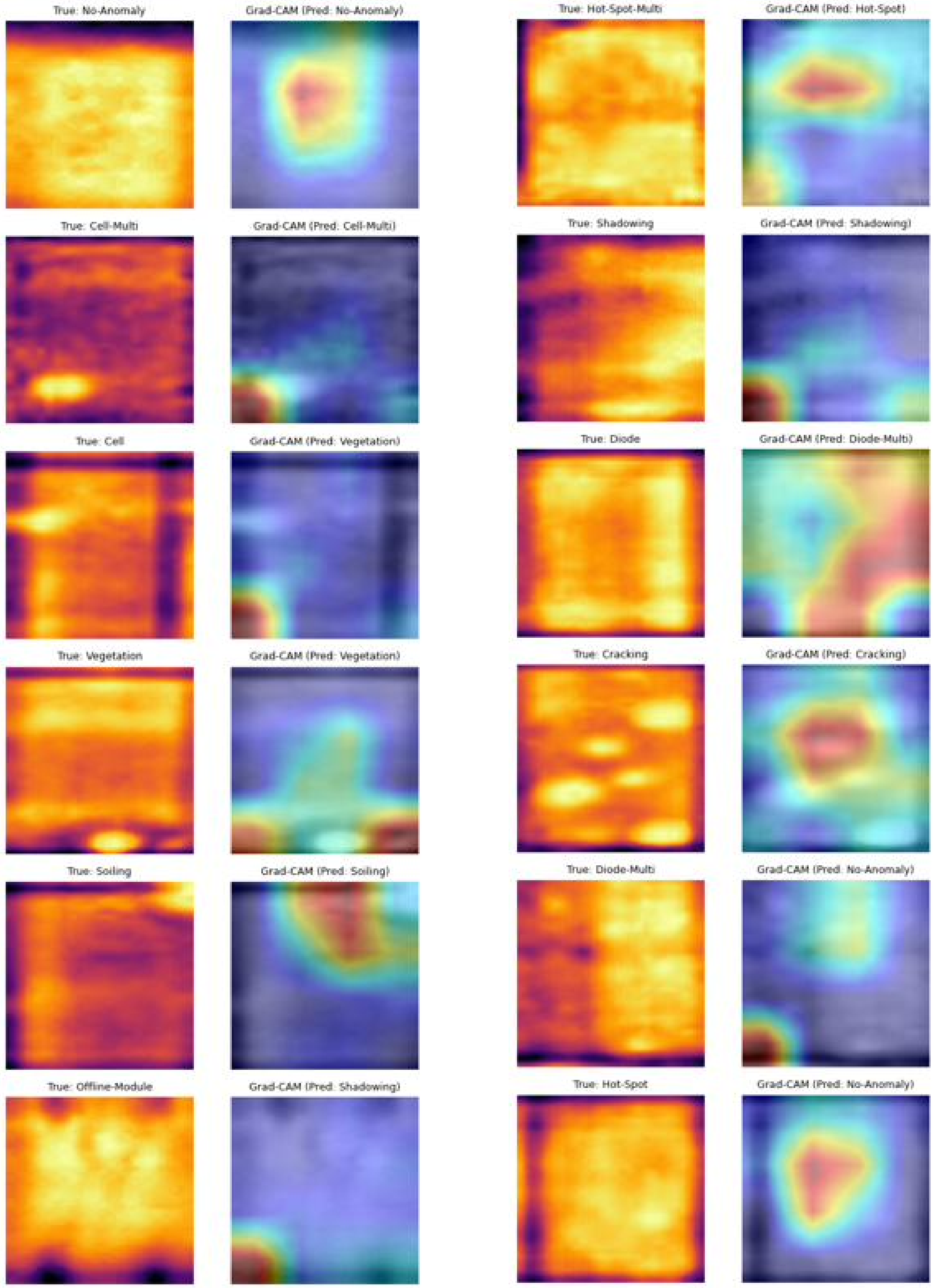

Figure 4.15: Grad-CAM heatmaps for representative images.

To look more closely at the rare fault classes, which are the hardest to classify and the most likely to be confused with the healthy class, Grad-CAM was applied to correctly classified examples of each rare fault. Figure 4.16 shows the result for Diode Multi, Hot Spot, Hot

Spot Multi, Soiling, and Cracking. In each case the bright region of the heatmap sits on the genuine defect, the hot region of a diode fault, the localised hot cell of a hot spot, or the affected area of a soiling or cracking case, rather than on the module frame or the background. This is important because it shows that the final model no longer treats these faint rare faults as ordinary healthy modules. The features that the backbone highlights for these classes are the real thermal signatures of the faults, which explains why the rare class scores improved in the final model and gives confidence that the improvement is genuine rather than an accident of the class balance.

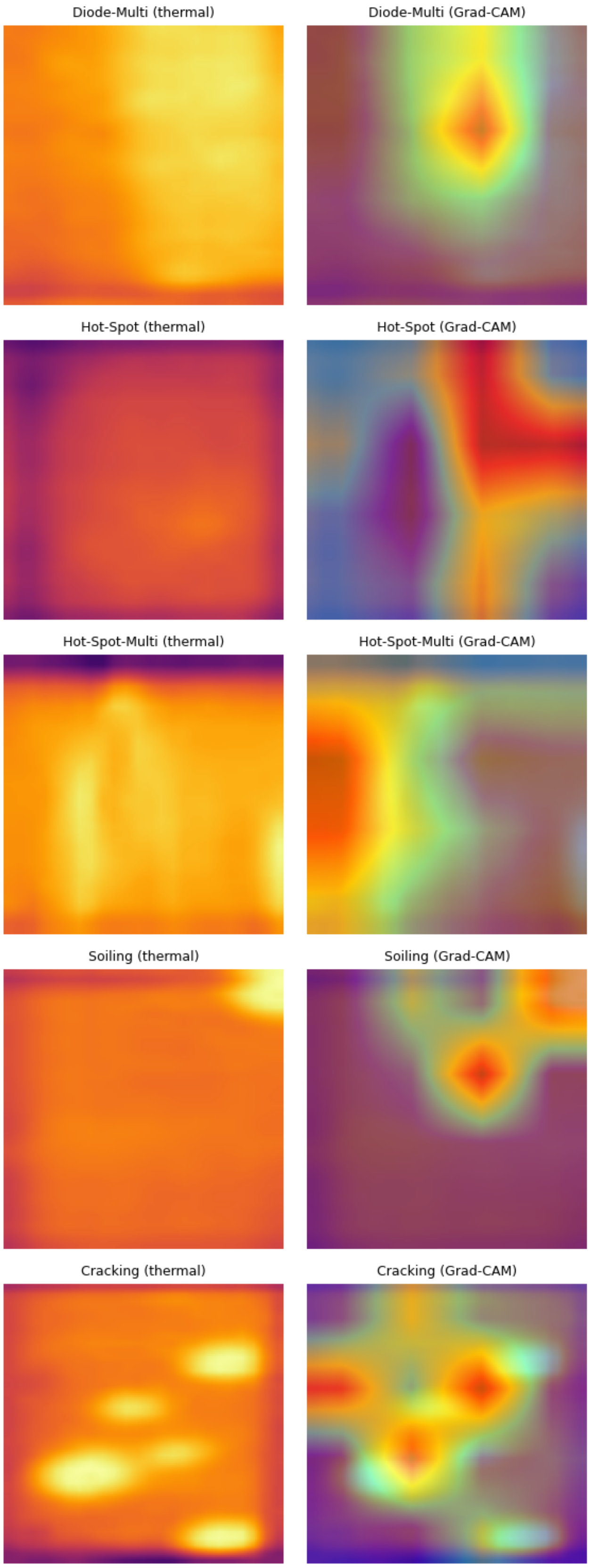

Figure 4.16: Grad-CAM heatmaps for correctly classified rare fault classes, showing that the model attends to the true defect region rather than confusing them with healthy modules.

4.7 Final Confusion Matrix

Figure 4.17 shows the confusion matrix of the final model, at 0.811 accuracy and 0.665 macro F1. The diagonal is strong, and the rare classes are handled well for their size. Most of the errors that remain are between classes that look similar at this low resolution, such as Cell, Cell Multi, and Vegetation, and between the faintest faults and the healthy class. These are understandable given how small the images are.

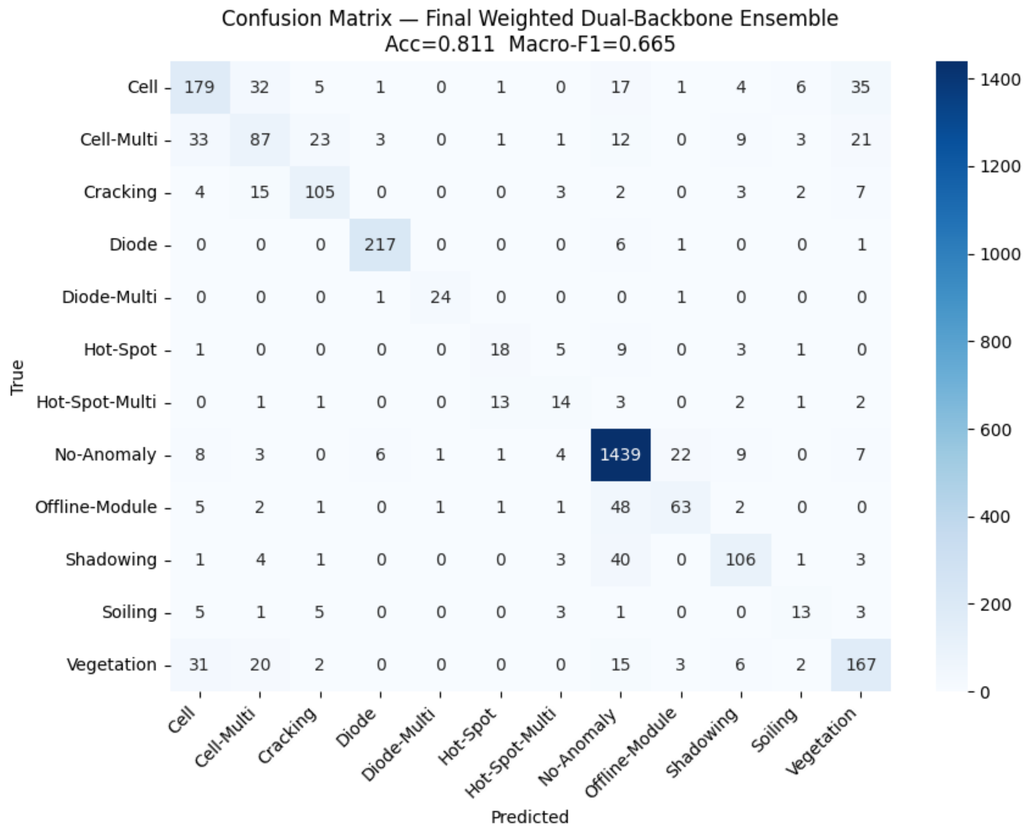

Figure 4.17: Confusion matrix of the final model, at 0.811 accuracy and 0.665 macro F1.

4.8 Summary

The results tell a clear story. On the binary task of finding faulty modules, the system is strong, with 0.930 accuracy and an area under the curve of 0.977. Used as a straight replacement for the raw image, the enhancement does not help, and it is worse on macro F1 in a multi seed test. But treated as a complement and fused with the raw image, then combined with balancing, fine tuning, and a two backbone ensemble, it contributes to a final result of 0.811 accuracy and 0.665 macro F1, with clear gains on the rare faults. The main contribution is both the method and the lesson behind it, that a preprocessing step should be judged by a fair test and used as a complement, not assumed to work as a substitute.

Chapter 5

Activities, Impacts and Sustainability

This chapter situates the technical work of the preceding chapters within its broader context. Section 5.1 describes the real-world problem the project addresses; Section 5.2 details the engineering activities carried out; Sections 5.3 and 5.4 examine the socio-economic and environmental impacts respectively; and Section 5.5 maps the work onto the relevant United Nations Sustainable Development Goals.

5.1 Relation with a Real-World Problem

The problem addressed by this thesis is drawn directly from the operations and maintenance of utility-scale solar photovoltaic (PV) plants. As established in Chapter 1, a single large plant contains tens or hundreds of thousands of individual modules, each of which can develop faults, cell cracking, bypass-diode activation, hot spots, soiling, shading, or complete module outages, that reduce energy yield, accelerate degradation, or, in the case of persistent hot spots, create a fire hazard. The economic and safety consequences of these faults scale with the size of the installation, and so does the difficulty of finding them.

Infrared thermographic inspection, increasingly performed by drone, has made it possible to acquire images of every module in a plant quickly. The unsolved part of the problem is interpretation: a human inspector cannot reliably and consistently review the resulting volume of thermal imagery, particularly for the rarer fault types that appear only occasionally. This is the specific, concrete gap that an automated classifier addresses, and it is the gap this thesis targets. The dataset used throughout the work, 20,000 thermal module crops labelled into the same twelve-class taxonomy used by commercial drone-inspection providers, is a direct representation of this industrial task rather than an artificial benchmark.

The work is therefore not an abstract exercise in image classification. Every design decision in the thesis traces back to a property of the real problem: the severe class imbalance reflects the fact that most modules in a real plant are healthy; the emphasis on macro-averaged F1 rather than raw accuracy reflects the operational reality that correctly flagging a rare, safety-critical fault matters more than correctly labelling yet another healthy module; and the interest in image enhancement reflects the genuinely low resolution and contrast of field-acquired thermal imagery.

5.2 Engineering Activities

Completing this thesis required a sequence of engineering activities spanning signal processing, machine learning, and rigorous experimental evaluation. These are summarised below in the order they were undertaken.

Data engineering: The InfraredSolarModules dataset was obtained, parsed from its metadata, and partitioned into training, validation, and test sets by stratified sampling that preserved the class distribution across all three splits. A strict check was performed to confirm no image overlap between splits, since leakage between training and test data would invalidate every subsequent result.

Signal processing and image enhancement: A band-selective hybrid wavelet enhancement pipeline was designed and implemented, comprising a two-dimensional discrete wavelet decomposition, contrast-limited adaptive histogram equalisation and gamma correction applied to the approximation sub-band only, a high-frequency gain applied to the detail sub-bands only, and an inverse wavelet reconstruction. This design itself resulted from an engineering diagnosis: an earlier whole-image cascade was found by ablation to degrade image quality, and the band-selective architecture was the corrective response.

Deep learning engineering: An EfficientNetV2-S backbone was adapted through two-phase transfer learning under the compute constraints of free cloud GPU environments. This required practical engineering, batching, memory management, and the extraction and caching of pooled feature embeddings, to keep the many required training runs within the available budget.

Experimental methodology: A controlled comparison between enhanced-input and raw-input models was designed, holding all other factors constant. Recognising the run-to-run variance inherent in training deep networks, the comparison was repeated across multiple random seeds and evaluated with a paired statistical test, rather than relying on a single run.

Class imbalance and feature fusion: Pooled features from the raw and enhanced representations were extracted, concatenated into a fused embedding, and balanced using the synthetic minority over-sampling technique. A range of classical classifiers and a soft-voting ensemble were then trained and compared on the fused representation to obtain the final result.

Explainability and validation: Grad-CAM was applied to confirm that the trained classifier attended to genuine defect regions, and per-class confusion analysis was used throughout to understand where and why misclassifications occurred.

Taken together, these activities exercise the full workflow of an applied machine learning engineering project: from raw data through a designed preprocessing stage, model training, statistically disciplined evaluation, and interpretation of results.

5.3 Socio-Economic Impact

Reliable automated PV fault classification carries socio-economic value along several distinct dimensions.

Reduced operations and maintenance cost: Automating the triage of thermal imagery reduces the number of skilled technician-hours required per megawatt of installed capacity. Because operations and maintenance is a recurring cost over a plant's multi-decade lifetime, even a modest reduction compounds substantially, and it lowers the levelised cost of the electricity the plant produces, a cost ultimately borne by consumers.

Protection of energy yield and financial return: Faults such as hot spots and diode failures directly reduce a module's output, and left undetected they erode the financial return on what is a capital-intensive investment. Earlier detection protects that return, and in the case of fire-risk faults it protects the physical asset and the surrounding installation as well.

Lower barriers for smaller operators: A low-cost automated screening tool reduces the expertise required to monitor an installation competently. This is particularly relevant for smaller commercial operators and for rural or off-grid solar deployments, which may lack access to specialised inspection services and for whom a single undetected fault represents a larger proportional loss.

A shift rather than a loss of skilled work: Automated screening does not remove the human role; it redirects it. Technician effort moves away from the repetitive visual review of thousands of near-identical thermal frames and toward higher-value tasks, physical inspection and repair, dispatch prioritisation, and verification of the system's outputs. It is worth being candid here about a limitation that bears directly on this point: at the accuracy levels obtained in this thesis, such a system is suited to assisting a human inspector by prioritising which modules warrant attention, not to replacing expert judgement outright.

5.4 Environmental Impact

The environmental benefits of this work follow directly from its effect on solar energy production and equipment longevity.

Improved effective solar yield: Faults such as soiling, shading, and cell-level degradation reduce the power a module delivers. Detecting and correcting them earlier keeps modules operating closer to their rated capacity, which increases the quantity of clean electricity generated per installed panel over its lifetime. Each additional unit of solar generation displaces generation that might otherwise come from fossil-fuel sources.

Extended equipment lifetime and reduced electronic waste: Catching faults before they cause irreversible damage, a hot spot before it permanently degrades a cell, for example, extends the service life of a module. Longer service life means fewer modules decommissioned prematurely, which reduces the volume of photovoltaic electronic waste. This matters because PV modules contain materials whose end-of-life disposal requires careful handling, and because manufacturing replacement modules itself carries an environmental cost.

More efficient inspection logistics: Where automated classification reduces the number of dedicated diagnostic site visits or inspection flights required to confirm and localise suspected faults, it correspondingly reduces the fuel and transport footprint associated with plant maintenance.

These benefits are real but should be stated in proportion: the contribution of any single inspection tool to overall emissions reduction is indirect and incremental, operating by helping the renewable-energy infrastructure function more efficiently rather than by reducing emissions directly.

5.5 Mapping with Sustainable Development Goals

The outcomes of this project align with several of the United Nations Sustainable Development Goals (SDGs). The most directly relevant are summarised in Table 5.1 and discussed below.

Table 5.1: Mapping of the project's contributions to the relevant United Nations Sustainable Development Goals.

SDG	Goal	Contribution of this work
7	Affordable and Clean Energy	Helps PV assets operate nearer to rated capacity for longer, supporting reliable and cost-effective renewable generation.
9	Industry, Innovation and Infrastructure	Applies modern deep learning and signal processing to build more intelligent, resilient infrastructure-monitoring tools.
12	Responsible Consumption and Production	Extends module service life through earlier fault detection, reducing premature replacement and electronic waste.
13	Climate Action	Supports more efficient solar generation, contributing indirectly to the displacement of fossil-fuel-based electricity.

SDG 7, Affordable and Clean Energy: By helping to keep solar assets operating near their rated capacity, and by lowering the maintenance cost that feeds into the levelised cost of electricity, the work supports the goal of increasing the availability and affordability of clean energy.

SDG 9, Industry, Innovation and Infrastructure: The project applies contemporary machine learning and image-processing techniques to a practical infrastructure-monitoring problem, contributing to the modernisation and resilience of renewable-energy infrastructure.

SDG 12, Responsible Consumption and Production: By enabling earlier fault detection and thereby extending module service life, the work helps reduce premature equipment replacement and the associated material consumption and electronic-waste burden.

SDG 13, Climate Action: By supporting more reliable and efficient solar generation, the work contributes, indirectly and at the margin, to the wider effort of replacing fossil-fuel-based generation with renewable sources.

In summary, although this thesis is at its core a focused technical study of image enhancement and classification, its motivation and its consequences are firmly connected to the real engineering problem of maintaining solar infrastructure, and through that problem to the broader goals of affordable clean energy and responsible, sustainable production.

Chapter 6

Conclusion

6.1 Concluding Remarks

This thesis set out to find whether a band selective hybrid wavelet enhancement improves the automated classification of photovoltaic module faults from thermal images, and to answer that question with more statistical care than is usual for this dataset. The work produced a clear answer, and along the way it found a better way of using the enhancement than the one first proposed.

The enhancement was designed with care. An earlier version that applied contrast adjustment and sharpening across the whole image was shown, by ablation, to reduce image quality rather than improve it, and this led to the corrected band selective design, in which contrast operations act only on the low frequency approximation band and sharpening acts only on the detail bands. Even with this careful design, the enhancement did not give the benefit that was expected. When used as a direct replacement for the raw image, and tested across three random seeds with a paired test, it did not reliably improve classification, and on the macro averaged F1 the raw image was significantly better. A single run would have hidden this, because two runs of the same setup disagreed on which input was better. The statistical test was therefore not a formality but the thing that made the result trustworthy.

The more useful finding grew out of taking that negative result seriously. Because the enhancement changes the image unevenly, helping some classes and harming others, the enhanced and raw views are not redundant, and this suggested combining them rather than choosing between them. Fusing the features of both views, correcting the class imbalance with SMOTE, fine tuning the backbone on the thermal images, and combining two different backbones in a weighted ensemble raised the twelve class accuracy from a raw baseline of 0.619 to 0.811, and the macro F1 from 0.481 to 0.665. On the binary task of separating healthy from faulty modules, the same system reached 0.930 accuracy and an area under the curve of 0.977. The main lesson is as much about method as about results: an enhancement that fails as a replacement can still carry useful information as a complement, and only a controlled evaluation tells the two cases apart.

6.2 Key Findings

The main findings of this thesis are as follows.

1. A band selective wavelet enhancement, which confines contrast operations to the approximation band and sharpening to the detail bands, is a sound correction to a whole image cascade that was shown by ablation to reduce image quality.

2. Used as a direct replacement for the raw image, the enhancement does not reliably improve classification, and it is significantly worse on macro F1 across three seeds, with a p value of 0.028. The accuracy difference is not significant.

3. Single run comparisons are unreliable on this task. Two runs of an identical setup gave better results on rare classes fault on Grad-Cam.

4. Treating the raw and enhanced views as complementary features, and combining them through fusion, SMOTE balancing, backbone fine tuning, and a two backbone weighted ensemble, gives the best twelve class result of 0.811 accuracy and 0.665 macro F1, with clear gains on the rare classes.

5. On the binary task of detecting whether a module is faulty at all, the system reaches 0.930 accuracy and an area under the curve of 0.977, catching the large majority of faults, which is the most useful result for practical field inspection.

6.3 Limitations

The findings should be read alongside the limits of the study. The native resolution of the images, at twenty four by forty pixels, sets a ceiling on how finely the classes can be separated, and it is the most likely reason that visually similar classes such as Cell, Cell Multi, and Vegetation stay hard to tell apart. The enhancement parameters were chosen by hand rather than optimized, and were kept the same for every class even though the ideal values differ by class. The class imbalance was only partly corrected, through weighting and oversampling, rather than through the full range of available methods. The results come from a single stratified test split, with several seeds used for the statistical comparison but not for full cross validation, so the absolute figures carry the normal uncertainty of a single split.

6.4 Future Work

Several of these limits point to useful next steps. The most promising is to fine tune both backbones fully and jointly on the thermal data, rather than using them mainly as feature extractors, which is the change most likely to raise absolute performance. A second direction is to optimize the enhancement parameters against the downstream macro F1 rather than an image quality score, and to allow different settings for different classes, since the per class analysis shows that no single setting suits every fault. A third is to strengthen the evaluation with k fold cross validation and, where possible, a cross plant test that measures how well the model carries over to installations it was not trained on, which is the condition that matters most for real deployment. Finally, the fusion used here was deliberately simple, and a learned or attention based fusion of the raw and enhanced branches, trained end to end, may make more of their complementary information than the plain concatenation used in this work. Beyond these direct steps, the broader contribution of the thesis, that a preprocessing method should be tested under controlled conditions and treated as a possible complement rather than only a substitute, applies to many other preprocessing methods in this field and offers a template for evaluating them more reliably than single run comparisons allow.

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Appendix A

Implementation Details

The system was implemented in Python. The wavelet decomposition used the PyWavelets library, and the contrast operations used OpenCV. The EfficientNetV2 backbone was implemented in TensorFlow and Keras, and the ConvNeXt backbone used PyTorch with the timm library. SMOTE was provided by the imbalanced learn library, and the classifiers, feature scaling, and metrics used scikit learn. Training was carried out on cloud graphics processing unit runtimes. The feature vectors from each backbone were computed once and cached to disk, so that the fusion, balancing, and ensemble stages could be repeated quickly without re running the costly backbone inference.

Appendix B

Class Labels and Test Support

Table B.1 lists the twelve classes of the dataset together with the number of samples in the test split, in the alphabetical order used throughout the thesis.

Table B.1: The twelve classes and their test sample counts.

Class	**Test samples**
Cell	281
Cell-Multi	193
Cracking	141
Diode	225
Diode-Multi	26
Hot-Spot	37
Hot-Spot-Multi	37
No-Anomaly	1500
Offline-Module	124
Shadowing	159
Soiling	31
Vegetation	246